Chassis Management for NVIDIA Switch Systems with Sysfs

User Manual

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What changed in this revision

Rev. 3.2.13, October 2026. Revisions through 3.2.12 are in the user manual changelog.

- Section references are links. The manual does not number its headings, so a reference such as “see Get Reset Cause” now points at that heading, and it stays valid when sections move.
- Drivers covers every kernel module and I2C device that the scripts load or instantiate: the PMBus regulators, the temperature sensors, the ADC, GPIO, EEPROM and I2C mux drivers, the host I2C controllers, and the generic `i2c-dev` and `ledtrig-timer` modules.

- Layout and rendering fixes: tables that showed raw markdown now render as text, the Get Reset Cause notes no longer run off the page, and section headings no longer sit alone at the bottom of a page.

Introduction

NVIDIA hw-management package uses a virtual file system provided by the Linux kernel called sysfs.

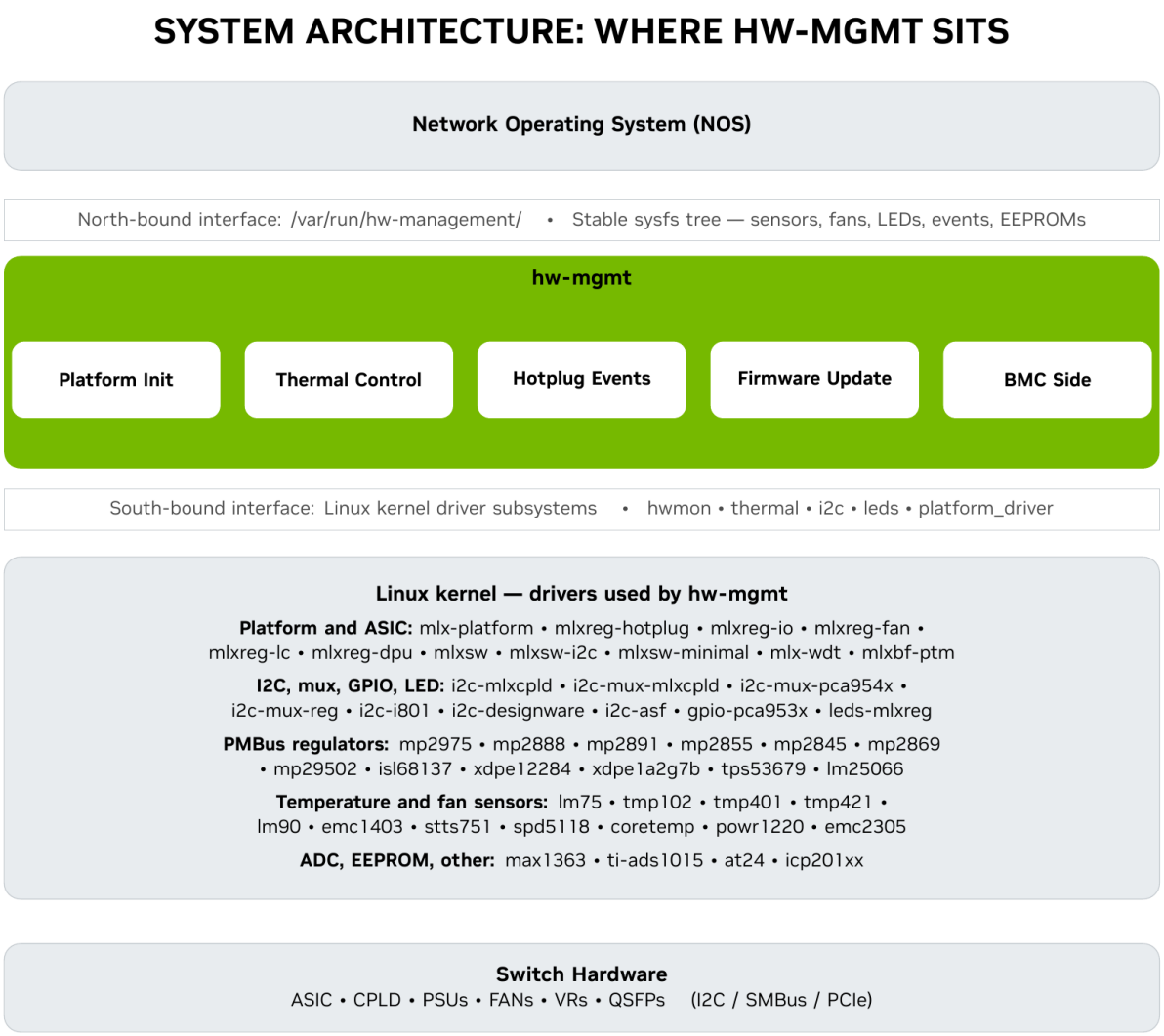
The sysfs file system enumerates the devices and buses attached to the system in a file system hierarchy that can be accessed from the user space.

The major advantage of working with sysfs is that it makes HW hierarchy easy to understand and control without having to learn about HW component location and the buses through which they are connected.

Software Components

Figure 1 presents the system architecture and the layer separation for sysfs support.

Figure 1 - System Architecture and Sysfs Layout



Host and BMC software stacks

The **hw-mgmt** repository ships two Debian packages that both expose the same virtual hierarchy under `/var/run/hw-management/` (`$bsp_path` in this manual), but they are built from **different source trees** and run on **different processors**:

	Host stack	BMC stack
Role	Switch CPU / NOS	Dedicated BMC SoC
Debian package	`hw-management`	`hw-management-bmc`
Repository source	`usr/` (scripts under `usr/usr/bin/`, platform data under `usr/etc/`)	`bmc/usr/` (mirrors host layout; platform data under `bmc/usr/etc/<HID>/`)
Installed on target	`/usr/bin/hw-management*.sh`, `/usr/bin/hw_management_thermal_control*.py`, `/lib/udev/rules.d/50-hw-management-events.rules`, …	`/usr/bin/hw-management-bmc*.sh`, `/lib/udev/rules.d/5-hw-management-bmc-events.rules`, `/etc/<HID>/`, …
Primary event handlers	`hw-management-chassis-events.sh`, `hw-management-thermal-events.sh`	`hw-management-bmc-events.sh` (and helpers under the same prefix)
Init / systemd	`hw-management.service`	`hw-management-bmc-init.service` and related BMC units (see `bmc/README.md`)
Examples / reference layouts	Platform sensors in `usr/etc/hw-management-sensors/`; scripts under `usr/usr/bin/`	Flat files under `bmc/examples/`; per-HID template at `bmc/examples/HIxxx/examples/`
Developer guide	Repository root `README.md`	`bmc/README.md`, `bmc/DEVELOPER_GUIDE.md`

How to use this manual

1. Find the node under `/var/run/hw-management`. The attribute sections call that directory `$bsp_path`.
2. Read **Access**. Read only means read the file. Read/write means the example includes the write.
3. Read the stack label when the section has one. **Host** is the package on the switch CPU. **BMC** is the package on the BMC. **Host + BMC** means the same name can exist on both images, with different handlers.
4. If the file is absent, this platform does not create that node.

5. Drivers describes the kernel modules under those nodes. The attribute sections are the names the NOS uses.

Both stacks may be present on the same product (CPU + BMC each running their own package); they do **not** share the same `/usr/bin` install tree on a single root filesystem.

Stack column (used in Virtual SysFS Hierarchy):

Label	Meaning
Host	Node created by `hw-management` on the switch CPU (`hw-management-thermal-events.sh`, `hw-management-chassis-events.sh`, `hw-management.sh`)
BMC	Node created by `hw-management-bmc` on the BMC SoC (`hw-management-bmc-events.sh` and related BMC scripts)
Host + BMC	Same logical name may exist on both images; paths/handlers differ — see section note

BMC stack — HI189 reference (`hw-management-bmc-events.sh`, validated against `bmc/usr/etc/HI189/5-hw-management-bmc-events.rules` and `hw-management-bmc-early-i2c-devices.json`):

`$bsp_path` subtree	Symlinks / behaviour	Peripheral (I2C / driver)	Stack
`thermal/cpu_temp_input`, `cpu_temp`, `cpu_min`	Host **CPU** temperature via BMC-side I2C	bus 15 / `0x4c`, **sbtsi**	BMC
`thermal/bmc_temp_input`, `bmc_temp`	**BMC board** ambient (`bmc_min` not created — **lm75** has no `temp1_min`)	bus 4 / `0x48`, **lm75**	BMC
`eeprom/eeprom_system`	System VPD EEPROM	bus 5 / `0x51`, **24c512**	BMC
`eeprom/eeprom_bmc`	BMC FRU EEPROM	bus 4 / `0x50`, **24c02**	BMC

$bsp_path subtree	Symlinks / behaviour	Peripheral (I2C / driver)	Stack
system/* (many attrs)	**mlxreg-io** / **mlxreg-hotplug** register files	CPLD / platform control on BMC	BMC
leakage/<N>/<j>/…	A2D leak-detector tree (input, thresholds, type)	ADS1015 / ADS7924 / MAX1363 per JSON config	BMC
bmc/reset_*, bmc/domains/reset_*, bmc/raw_scu*	AST2700 BMC reset-cause exporter	hw-management-bmc-get-reset-cause.sh	BMC

Example layouts: **bmc/examples/hw-management-bmc-thermal-sysfs.txt** (delivered path), bmc/examples/hw-management-bmc-eeprom-sysfs.txt, bmc/examples/hw-management-bmc-system-sysfs.txt, bmc/examples/hw-management-bmc-leakage-sysfs.txt.

Host stack — BMC-related nodes (CPU image, not the BMC package):

Node	Evidence	Stack
system/bmc_present	GPIO symlink (host hw-management.sh platform GPIO mapping)	Host
system/bmc_to_cpu_ctrl	**mlxreg-io** on CPU (mlxplat; same tree)	Host
system/cpu_mctp_ready	**mlxreg-io** on CPU (same tree)	Host
config/mctp_addr, config/mctp_bus	Written by hw-management.sh when platform defines MCTP (e.g. N5110)	Host

Do not assume host ASIC/PSU/fan thermal nodes exist on the BMC image, or that BMC cpu_temp_* nodes exist on the CPU image — check $bsp_path on the target root filesystem.

Virtual / SimX platforms (no production sysfs tree in repo): HI194 (N7200_LD), HI199 (N7300_LD), HI200 (N7400_LD) and other SimX SKUs may exit early from hw-management-

`ready.sh` with mock values only (`check_simx` paths in `hw-management.sh`). Do not use these for production attribute validation.

Hierarchy and Structure

The package uses the Linux default hierarchy structure of sysfs under the directory /var/run/hw-management.

This path is used by existing applications that use auto-discovery to find existing HW components. Two examples for such applications are:

- libsysfs – the libraries provide a consistent and stable interface for querying system device information exposed through sysfs.
- systool – a utility built upon libsysfs that lists devices by bus, class, and topology.

The disadvantage of using this path is that the hierarchy model includes the BUS type and location model which is subject to change between different system types.

To resolve this limitation, the virtual hierarchy structure that is not HW dependent is supported. This hierarchy is a collection of soft links to the default sysfs structure. This document describes the way to work with this hierarchy in order to control the HW.

Chassis attributes information exported through sysfs can be utilized by a number of standard Linux tools. So, for example, the following are tools from the Linux packages lm-sensors and fancontrol, which are capable of operating on top of sysfs infrastructure:

- pwmconfig – tests the pulse width modulation (PWM) outputs of sensors and configures fancontrol
- fancontrol – automated software-based fan speed regulation
- sensors – print sensors information

Sysfs Initialization and Driver Registration

As described in the previous sections, sysfs structure provides access to HW drivers. These drivers need to be initialized before using sysfs. In addition, NVIDIA virtual hierarchy also needs to be created in order to use it.

The following applies to the **host** package (`hw-management`). For the **BMC** package, see Host and BMC software stacks and `bmc/README.md` (systemd units, udev rules, and scripts under `hw-management-bmc-*`).

The package provides a simple way to initialize the drivers using the set of shell scripts. These scripts support initialization and de-initialization of driver, virtual hierarchy structure, udev events handling, based on a set of NVIDIA system specific udev rules.

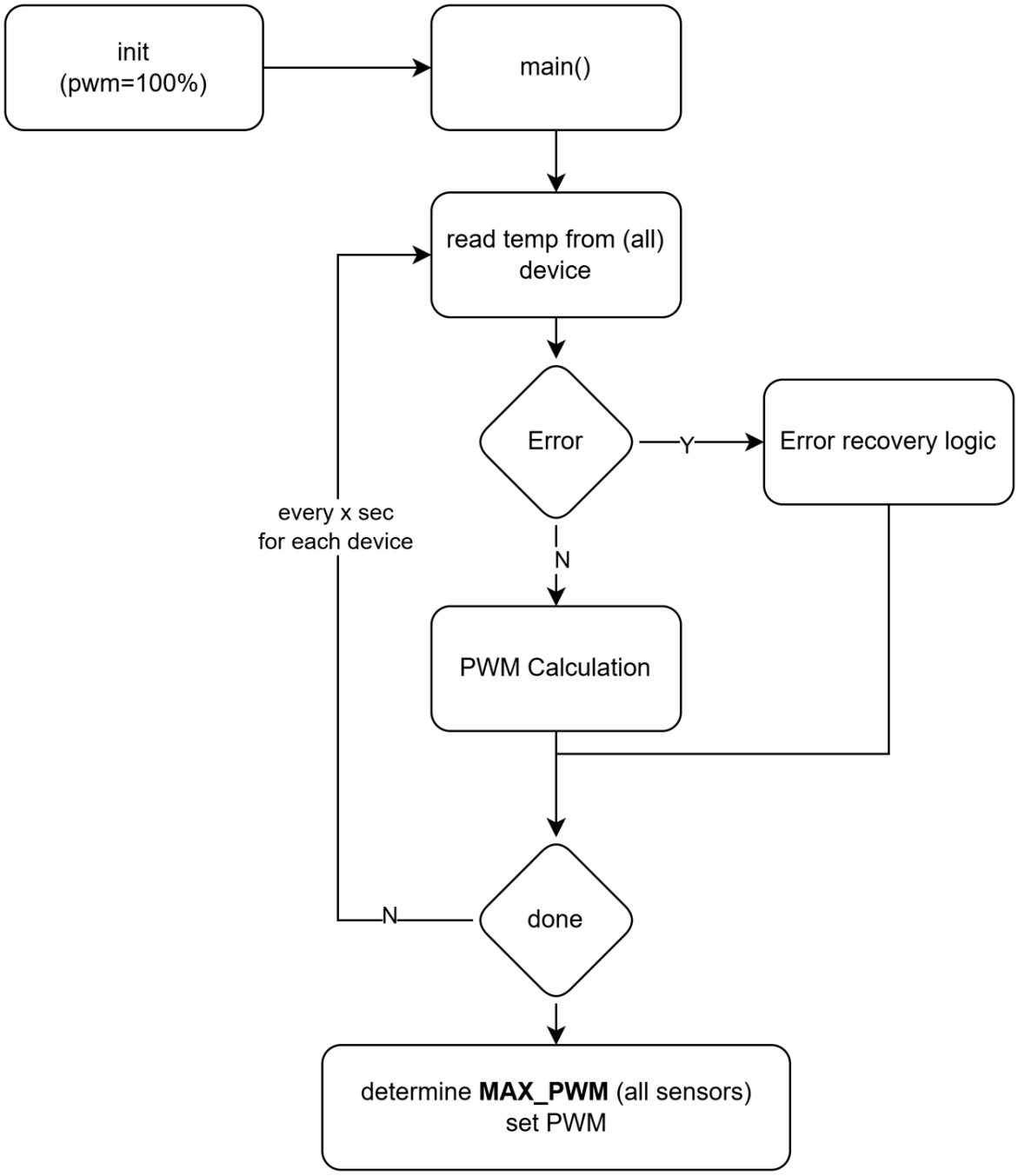
Package contains the following files, used within the workload:

- /lib/systemd/system/hw-management.service: system entries for thermal control activation and de-activation.
- /lib/udev/rules.d/50-hw-management-events.rules: udev rules defining the triggers on which events should be handled. When trigger is matched, rule data is to be passed to the event handler (see below file /usr/bin/hw-management-events.sh).
- /usr/bin/hw_management_thermal_control.py: contains thermal algorithm implementation (TC v2.0). /usr/bin/hw_management_thermal_control_2_5.py: contains thermal algorithm implementation (TC v2.5).
- /usr/bin/hw-management-chassis-events.sh and /usr/bin/hw-management-thermal-events.sh: handle udev triggers, according to the received data, it creates or destroys symbolic links to sysfs entries. It allows to create system independent entries, and it allows thermal controls to work over this system independent model. Raises signal to thermal control service in case of fast temperature decreasing. It could happen in case one or few very hot port cables have been removed. Sets PS units internal FAN speed to default value when unit is connected to power source.
- /usr/bin/hw-management.sh: performs initialization and de-initialization, detects the system type, connects thermal drivers according to the system topology, activates and deactivates thermal algorithm.
- /usr/bin/hw-management-led-state-conversion.sh and /usr/bin/hw-management-power- helper.sh: helper scripts.

- /etc/modprobe.d/hw-management.conf and /etc/modules-load.d/hw-management-modules.conf: configuration for kernel modules loading.

For more details follow package README file (`README.md` for host, `bmc/README.md` for BMC).

Figure 2 - Thermal Management Flow



Virtual SysFS Hierarchy

NVIDIA virtual hierarchy supports the following HW control ($bsp_path below is a location of virtual SysFS hierarchy, in standard Linux distributions, like Debian, RedHat, Fedora, etcetera this is /var/run/hw-management folder).

Table 1 - NVIDIA Hierarchy Node Support

Node Path	Purpose
$bsp_path/config	Internal system specific configuration data
$bsp_path/eeprom	Gets raw data from EEPROM in system modules
$bsp_path/environment	Gets information on environmental sensors (A2D, Volt, Curr)
$bsp_path/led	Gets/sets LED color (see **LED_Control_API.md** for runtime `led_<func>` naming and CPLD behaviour)
$bsp_path/bin	
$bsp_path/events	
$bsp_path/firmware	
$bsp_path/ui	Textual labels for sensors
$bsp_path/power	Gets information from power sensors
$bsp_path/system	Gets/sets system variables and settings (CPLD version, fan dir, reset, pwr cycle)
$bsp_path/thermal	Gets variant thermal sensors in systems and gets/sets fan attributes

Node Path	Purpose
$bsp_path/watchdog	Standard watchdog sysfs attributes
$bsp_path/alarm	Get System chassis
$bsp_path/jtag	Provides interface for JTAG CPLD burn
$bsp_path/sysfs_labels_rdy	
$bsp_path/fast_sysfs_labels_rdy	
$bsp_path/sysfs_mlxsw_rdy	Uptime in milliseconds after the mlxsw hwmon links are created. SPC1 indication; the script keys off hwmon name `mlxsw`.

Detailed information on each of these nodes can be found in the following sections.

Stack applicability: Most nodes in Virtual SysFS Hierarchy are created by the **host** stack (`hw-management-thermal-events.sh` / `hw-management-chassis-events.sh`). Sections tagged **Stack: BMC** or naming **`hw-management-bmc-events.sh`** apply to the **BMC package** only (see Host and BMC software stacks). BMC status documents host-visible BMC status; BMC `system/` register layout is in `bmc/examples/hw-management-bmc-system-sysfs.txt`.

Note: some of the attributes described below are not relevant to all platforms and will exist only on the platforms which support this attribute.

mlxsw sysfs ready

Node name	`$bsp_path/sysfs_mlxsw_rdy`
Description	Written by `hw-management-thermal-events.sh` on the switch add path when the hwmon `name` file is `mlxsw`, after that hwmon's module temperature links are created. The value is `/proc/uptime` converted to milliseconds (`awk '{print int($1 * 1000)}'`). The helper comment describes the file as the module, ASIC, fan, and PWM ready indication for SPC1. The script does not test a SKU; it tests the hwmon name. The tree writes this file and does not delete it. Introduced by commit `8775d0d3` (bug 5207348), listed in the V.7.0070.1932 release notes.
Access	Read only
Release version	V.7.0070.1932
Example	Read the ready timestamp: `cat $bsp_path/sysfs_mlxsw_rdy`

Config Control

Get ASIC Bus

Node name	`$bsp_path/config/asic_bus`
Description	Get system ASIC bus number
Values	Status, Integer: 1-99
Access	Read only
Release version	1.0
Example	Get asic bus number: `cat $bsp_path/config/asic_bus`

Get ASIC I2C Bus

Node name	`$bsp_path/config/asic<index>_i2c_bus_id`
Description	Get ASIC <index> I2C bus
Access	Read only
Release version	V.7.0030.2930
Example	Get ASIC I2C bus address: `cat $bsp_path/config/asic1_i2c_bus_id`

Get ASIC PCI Bus

Node name	`$bsp_path/config/asic<index>_pci_bus_id`
Description	Get ASIC PCI bus ID
Access	Read only
Release version	V.7.0020.1338
Example	Get asic PCI bus number: `cat $bsp_path/config/asic1_pci_bus_id`

Get ASIC Chip-up completed

Node name	`$bsp_path/config/asic_chipup_completed`
Description	Get ASIC count which has already initialized. When asic_chipup_completed == asic_num, asics_init_done should be set to ”1’
Values	Status, Integer: 0..asic_count
Access	Read
Release version	1.0
Example	Get already initialized asic count: `cat $bsp_path/config/asic_chipup_completed`

Get ASIC init done

Node name	`$bsp_path/config/asics_init_done`
Description	Get ASIC init done status. 1 – All asic initialized and ready. 0 – One or more ASICs not ready When asic_chipup_completed == asic_num, asics_init_done should be set to "1'
Values	Status, Integer: 1 – ALL ASICs ready; 0 – One or more ASICs not ready
Access	Read
Release version	1.0
Example	Get asics init ready: `cat $bsp_path/config/asics_init_done`

Set Chip-down/Chip-up Delay

Node name	`$bsp_path/config/chipdown_delay` `$bsp_path/config/chipup_delay`		
Description	Set delay duration in seconds for hw mgmt service from the chip down/up event.		
Access	Write/Read		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	Status	Integer (seconds)	0 – no delay other – delay
Example	Get chipdown value: `cat $bsp_path/config/chipdown_delay` Set 5 seconds delay in chipup value: `echo 5 > $bsp_path/config/chipup_delay`		

Read CPLD Number

Node name	`$bsp_path/config/cpld_num`
Description	Get the number of CPLD modules in the system
Values	Status, Integer: 1-X
Access	Read only
Release version	1.0
Example	Get CPLD number: `cat $bsp_path/config/cpld_num`

Read PSU VPD Info

Node name	`$bsp_path/eeprom/psu{n}_vpd`
Description	Get PSU VPD info in human readable format
Values	Status, ASCII: EEPROM info
Access	Read only
Release version	V.7.0010.1300
Example	Get PSU VPD info: `cat $bsp_path/eeprom/psu{n}_vpd`

Get Hot-plug Fan Number

Node name	`$bsp_path/config/hotplug_fans`
Description	Get hot-plug FAN number in the system
Values	Status, Integer: 0-X
Access	Read only It can be zero on fixed system.
Release version	V.7.0010.1300
Example	Get hot-plug fan number: `cat $bsp_path/config/hotplug_fans`

Get Hot-plug PSU Number

Node name	`$bsp_path/config/hotplug_psus`
Description	Get hot-plug PSU number in the system. It can be zero on fixed system.
Values	Status, Integer: 0-X
Access	Read only
Release version	V.7.0010.1300
Example	Get hot-plug psu number: `cat $bsp_path/config/hotplug_psus`

Get Hot-plug PDB Number

Node name	`$bsp_path/config/hotplug_pdbs`
Description	Get the number of hot-pluggable Power Distribution Boards (PDB) in the system. Note: This attribute is primarily for liquid-cooled systems (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD, N6300_LD; SN66XX_LD family: SN6600_LD). PDBs manage power distribution in liquid-cooled systems where traditional PSUs are not present. It can be zero on air-cooled systems or systems without hot-pluggable PDBs. Note: N6100_LD has hotplug_pdbs=0 (PDB is not hot-pluggable). Note: N6300_LD (SKU HI185) has hotplug_pdbs=2. Note: SN6600_LD (SKU HI193) has hotplug_pdbs=2 with `events/pdb1` and `events/pdb2`.
Values	Status, Integer: 0-X (number of hot-pluggable PDBs)
Access	Read only
Release version	V.7.0040.3930
Example	Get hot-plug PDB number: `cat $bsp_path/config/hotplug_pdbs`

Get Hot-plug PWR Number

Node name	`$bsp_path/config/hotplug_pwrs`
Description	Get hot-plug Power cable number in the system. It can be zero on fixed system.
Values	Status, Integer: 0-X
Example	Get hot-plug Power cable number: `cat $bsp_path/config/hotplug_pwrs`
Access	Read only
Release version	V.7.0010.1300

Get Hot-plug Linecards

Node name	`$bsp_path/config/hotplug_linecards`
Description	Get the number of Linecards
Access	Read only
Release version	
Example	Get the number of hot-plug linecards: `cat $bsp_path/config/hotplug_linecards`

Read Gearbox Counter

Node name	`$bsp_path/config/gearbox_counter`
Description	Get the number of gearboxes in the system.
Values	Integer: 0-X
Access	Read only
Release version	V.7.0010.3000
Example	`cat $bsp_path/config/gearbox_counter`

Read Module Counter

Node name	`$bsp_path/config/module_counter`
Description	Get the number of sfp modules in the system Note: this attribute is valid only for I2C ASIC driver
Values	Status, Integer: 1-X
Access	Read only
Release version	1.0
Example	Get sfp module: `cat $bsp_path/config/module_counter`

Read ASIC Chipup Counter

Node name	`$bsp_path/config/asic_chipup_counter`
Description	Number of remaining ASIC chip-up retry attempts. This counter is decremented with each chip-up attempt and reset to the initial retry count when all attempts are exhausted.
Access	Read only
Release version	3.0
Example	Get asic chipup completed: `cat $bsp_path/config/asic_chipup_counter`

Read ASIC Chipup Completed

Node name	`$bsp_path/config/asic_chipup_completed`
Description	counter of successful ASIC driver initialization completions: 0 - no successful initialization completion. 1 - one ASIC device has been successful initialized. n – 'n' ASIC devices has been successful initialized.
Values	Status, Integer: 1-X
Access	Read only
Release version	1.0
Example	Get asic chipup completed: `cat $bsp_path/config/asic_chipup_completed`

Read Init Done

Node name	`$bsp_path/config/asics_init_done`
Description	is to be set to one, when 'asic_chipup_completed' attribute matches 'asic_num' attribute (old static attribute /var/run/hw-management/config/asic_num)
Values	Status, Integer: 1-X
Access	Read only
Release version	1.0
Example	Get asics init done: `cat $bsp_path/config/asics_init_done`

Read Max System Fans (rotors)

Node name	`$bsp_path/config/max_tachos`
Description	Get max number of system fans.
Values	Status, Integer: 1-X
Access	Read only
Release version	1.0
Example	Get fans max value: `cat $bsp_path/config/max_tachos`

Read Fan Drawer Number

Node name	`$bsp_path/config/fan_drwr_num`
Description	Get number of system FAN drawers
Values	Status, Integer: 1-X
Access	Read only
Release version	1.0
Example	Get number of system FAN drawers: `cat $bsp_path/config/fan_drwr_num`

Read Fan Command

Node name	`$bsp_path/config/fan_command`
Description	Get PMBUS command for PSU config
Values	Status, Hex: 0xhh
Access	Read only
Release version	1.0
Example	Get fan command: `cat $bsp_path/config/fan_command`

Read Fan Max/Min Speed

Node name	`$bsp_path/config/fan_max_speed` `$bsp_path/config/fan_min_speed`
Description	Get the absolute system fan max/min speed
Values	Status, Integer: X
Access	Read only
Release version	1.0
Example	Get fan max speed: `cat $bsp_path/config/fan_max_speed` Get fan min speed: `cat $bsp_path/config/fan_min_speed`

Read Fan Max/Min Speed for front/rear FAN

Node name	`$bsp_path/config/fan_front_max_speed`
	`$bsp_path/config/fan_front_min_speed`
	`$bsp_path/config/fan_rear_max_speed`
	`$bsp_path/config/fan_rear_min_speed`

Description	Get the absolute system fan max/min speed for front/rear FAN. These attributes can be present on some switch types. If not present - fan_max_speed/fan_min_speed should be used instead.
Values	Status, Integer: X
Access	Read only
Release version	1.0
Example	Get front fan max speed: `cat $bsp_path/config/fan_front_max_speed` Get front fan min speed: `cat $bsp_path/config/fan_front_min_speed`

Fan Speed Tolerance

Node name	`$bsp_path/config/fan_speed_tolerance`
Description	Max tolerance for measured FAN min/max speed compared to reference defined in fan_max_speed/fan_min_speed
Values	int (percent): 0..90 (default 15)
Access	Read only
Release version	V.7.0040.1032
Example	`cat $bsp_path/config/fan_speed_tolerance`

Fan Speed Units

Node name	`$bsp_path/config/fan_speed_units`
Description	Value to write PSU PMBUS register FAN_CONFIG_1_2 Set FAN Commanded in Duty Cycle. Used for PSU fan SET configuration
Values	Status, Integer: X (default 0x90)
Access	Read only
Release version	V.7.10.3000
Example	`cat $bsp_path/config/fan_speed_units i2cset -f -y "$bus" "$addr" "$fan_config_command" "$fan_speed_units" bp`

Number of Leakage Sensors

Node name	`$bsp_path/config/leakage_counter`
Description	Get the number of leakage sensors installed in the switch.
Values	Status, Integer: 1-8
Access	RO
Release version	3.0
Example	`cat $bsp_path/config/leakage_counter` Note: On SN6600_LD (SKU HI193), `sn66xxld_specific()` in `hw-management.sh` sets `leakage_count` to **2**, which becomes `config/leakage_counter`. The same script initializes `events/leakage1` and `events/leakage2` only. Additional `system/leakage<N>` symlinks (for example `leakage3` through `leakage5` in the validated tree) can appear when mlxreg-io exposes those hwmon attributes; `hw-management-thermal-events.sh` links each present `leakageN` up to its `max_leakage` limit (**8**). Consumers should treat `leakage_counter` as the supported event and policy count unless platform documentation states otherwise.

Number of Cable Cartridges

Node name	`$bsp_path/config/cartridge_counter`
Description	Get the number of cable cartridges in the system. Cable cartridges are used in liquid-cooled multi-ASIC systems for connecting external cables. Note: This attribute is primarily for liquid-cooled systems with cable cartridges (N61XX_LD family: N6100_LD, N6300_LD). N6100_LD and N6300_LD (SKU HI185) have 4 cable cartridges (`config/cartridge_counter` = 4).
Values	Count, Integer: 0-X (number of cartridges)
Access	Read only
Release version	V.7.0060.1000
Example	Get cartridge count: `cat $bsp_path/config/cartridge_counter`

Read/write Time Window for Thermal Control Periodic Log Report

Node name	`$bsp_path/config/periodic_report`		
Description	Get/Set time for thermal control periodic log report (sec, default 7200)		
Access	Read/Write		
Release version	V.7.0010.1300		
Arguments	**Name**	**Data type**	**Values**
	Status	Integer	X
Example	Set periodic log report time: `echo 3000 > $bsp_path/config/periodic_report`		

Read PSU I2C Address

Node name	`$bsp_path/config/psu<power supply module number>_i2c_addr`
Description	Get the I2C address of PSU for direct connection
Values	Status, Hex: 0xhh
Access	Read only
Release version	1.0
Example	Get PSU1 I2C address: `cat $bsp_path/config/psu1_i2c_addr`

Read PSU I2C Bus

Node name	`$bsp_path/config/psu<X>_i2c_bus` Where <X> is the power supply module index (for example `psu1_i2c_bus`, `psu2_i2c_bus`).
Description	Contains the I2C bus number of PSU<X> used for direct connection to that power supply module. This configuration parameter is **optional** and is **not mandatory** on all platforms. It is created and used by the hw-management package for internal purposes (for example dummy PSU detection and mapping on supported switch systems). The OS and user-space applications **must not** assume that `psu<X>_i2c_bus` is always present; check for the file before reading it.
Values	Bus index, Integer: Platform-specific I2C bus number (for example 3, 4)
Access	Read only **Mandatory:** No
Release version	3.2
Example	Get PSU1 I2C bus when the file exists: `if [ -f $bsp_path/config/psu1_i2c_bus ]; then cat $bsp_path/config/psu1_i2c_bus fi`

Read Thermal Delay

Node name	`$bsp_path/config/thermal_delay`
Description	Get the delay duration (seconds) since the HW mgmt service starts until thermal control init
Values	Status, Integer (seconds): X
Access	Read only
Release version	1.0
Example	Get thermal delay: `cat $bsp_path/config/thermal_delay`

Dummy PSUs Supported

Node name	`$bsp_path/config/dummy_psus_supported`
Description	Indicates whether the system supports dummy PSUs for power management. Set to "1" for systems that support dummy PSU functionality, "0" otherwise.
Values	Integer: 0 or 1
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/dummy_psus_supported 1`

Read PSU power capacity

Node name	`$bsp_path/config/psu[x]_power_capacity`
Description	Get the maximum power capacity for the psu.
Values	Constant, Milli watt: X
Access	Read only This attribute is present only in SN5600 and SN5400 platforms.
Release version	1.0
Example	Get power capacity of the psu `cat $bsp_path/config/psu[x]_power_capacity`

Read PSU power slope

Node name	`$bsp_path/config/psu[x]_power_slope`
Description	Get the power slope value for the psu. This is a hardware defined constant and will be used in power calculation
Values	Constant, Integer: X
Access	Read only This attribute is present only in SN5600 and SN5400 platforms.
Release version	1.0
Example	Get power slope for the psu `cat $bsp_path/config/psu[x]_power_slope`

Read DPU Number

Node name	`$bsp_path/config/dpu_num`
Description	Number of DPUs (1-4)
Values	Status, Integer (number): X
Access	Read only
Release version	1.0
Example	Get dpu number: `cat $bsp_path/config/dpu_num`

Read DPU Board type

Node name	`$bsp_path/config/dpu_board_type`
Description	Whether DPU sensors are loaded static or dynamic
Values	Status, String: Static / dynamic
Access	Read only
Release version	1.0
Example	Get dpu board type: `cat $bsp_path/config/dpu_board_type`

Read DPU board bus offset

Node name	`$bsp_path/config/dpu_brd_bus_offset`
Description	DPU i2c bus offset
Values	Status, Integer (number): X
Access	Read only
Release version	1.0
Example	Get dpu i2c bus offset `cat $bsp_path/config/dpu_brd_bus_offset`

Read DPU bus offset

Node name	`$bsp_path/config/dpu_bus_off`
Description	I2c bus offset for the dpu
Values	Status, Integer (number): X
Access	Read only
Release version	1.0
Example	Get dpu bus offset number: `cat $bsp_path/config/dpu_bus_off`

Read DPU events

Node name	`$bsp_path/config/dpu_events`
Description	Events supported by DPUs
Values	Status, String: Ev1, ev2, etc.
Access	Read only
Release version	1.0
Example	Get the events supported by DPU `cat $bsp_path/config/dpu_events`

Read DPU events to host

Node name	`$bsp_path/config/dpu_to_host_events`
Description	DPU events to host
Values	Status, String: X
Access	Read only
Release version	1.0
Example	Get dpu events to host `cat $bsp_path/config/dpu_to_host_events`

Labels Ready

Node name	`$bsp_path/config/labels_ready`
Description	Label folder $bsp_path/ui_tree ready to use
Values	0 – labels init in progress; 1 – labels init ready
Access	Read only
Release version	V.7.0030.0958
Example	Read the node: `cat $bsp_path/config/labels_ready`

CPU Type

Node name	`$bsp_path/config/cpu_type`
Description	CPU type ID. Extracted from /proc/cpuinfo. Format: 0xXXYY XX – model num YY - cpu family
Values	Hex
Access	Read Only
Release version	
Example	`cat $bsp_path/config/cpu_type 0x656 cpu family : 6 model : 86 model name : Intel(R) Xeon(R) CPU D-1527 @ 2.20GHz`

Named Busses

Node name	`$bsp_path/config/named_busses`
Description	List of I2C bus idx/bus names separated by 'space'. Contains meaningful names for I2C busses on main board Present on some systems.
Access	Read Only
Release version	V.7.0020.3100
Example	asic1 2 pwr 4 vr1 5 amb1 7 vpd 8

I2C Bus Offset

Node name	`$bsp_path/config/i2c_bus_offset`
Description	Base i2c bus idx. used for internal purposes
Access	Read only
Release version	
Example	`cat $bsp_path/config/i2c_bus_offset 2`

I2C Bus Connect Devices

Node name	`$bsp_path/config/i2c_bus_connect_devices`
Description	List of i2c devices/names/bus/add used for internal purposes
Access	Read only
Release version	
Example	xdpe12284 0x62 5 voltmon1 xdpe12284 0x64 5 voltmon2 ...

I2C Bus Default Off EEPROM CPU

Node name	`$bsp_path/config/i2c_bus_def_off_eeprom_cpu`
Description	Offset of i2c bus for CPU VPD eeprom
Access	Read only
Release version	
Example	Read the node: `cat $bsp_path/config/i2c_bus_def_off_eeprom_cpu`

I2C Comex Mon Bus Default

Node name	`$bsp_path/config/i2c_comex_mon_bus_default`
Description	Default I2C bus number for COMEX monitoring devices. Used for voltage monitoring and other COMEX-related sensor readings.
Values	Integer: Bus number
Access	Read only
Release version	3.0
Example	Read the node: `cat $bsp_path/config/i2c_comex_mon_bus_default`

I2C SWB Bus

Node name	`$bsp_path/config/i2c_swb_bus`
Description	Absolute I2C adapter number for the switch-board (SWB) CPLD used for cartridge identity registers (rack id, topology id, tray id, slot id) at slave address `0x31`. Platform-specific (for example HI176 CPU bus 18, HI180 CPU bus 53). Created from platform.json `variables.i2c_swb_bus` or from legacy `*_specific()` init. Used by `hw-management-generate-dump.sh` to produce archive member `cpld_swb_cartridge_dump` when the node exists (rack_id 13 bytes + ASCII, topology/tray/slot).
Values	Integer: Bus number
Access	Read only
Release version	
Example	`cat $bsp_path/config/i2c_swb_bus 53` Example `cpld_swb_cartridge_dump` (from `/tmp/hw-mgmt-dump.tar.gz`): “`text i2c_swb_bus=53 addr=0x31 rack_id: 0x31 0x38 0x32 0x34 0x32 0x32 0x35 0x34 0x31 0x30 0x30 0x31 0x33 rack_id_ascii: 1824225410013 topology_id: 0x00 tray_id: 0x00 slot_id: 0x01 “`

LM Sensors Configuration

Node name	`$bsp_path/config/lm_sensors_config`
Description	Configuration file for sensor tool from lm_sensors package. Contains sensors definition for the system
Values	Text file
Access	Read only
Release version	
Example	`sensors -c $bsp_path/config/lm_sensors_config`

LM Sensor Labels

Node name	`$bsp_path/config/lm_sensors_labels`
Description	Path to JSON file containing sensor labels for lm-sensors configuration. Provides human-readable names for various temperature, voltage, and fan sensors.
Values	Text file
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/lm_sensors_labels /etc/hw-management-sensors/msn2700_sensors_labels.json`

Events Ready

Node name	`$bsp_path/config/events_ready`
Description	Provides indication that VPD parsing was completed.
Values	Status, Integer: 0/1
Access	Read only
Release version	3.0
Example	Check whether VPD parsing has been completed `cat $bsp_path/config/events_ready`

Minimal Driver Unsupported

Node name	`$bsp_path/config/minimal_unsupported`
Description	Provides indication for whether ASIC I2C ('minimal') driver is supported.
Values	Status, Integer: 0/1
Access	Read Only
Release version	3.0
Example	Get indication on whether minimal driver is supported: `$bsp_path/config/minimal_unsupported`

SGMII PHY

Node name	`$bsp_path/system/sgmii_phy`
Description	SGMII (Serial Gigabit Media Independent Interface) PHY status and configuration. Provides information about the physical layer interface for gigabit Ethernet connections.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/system/sgmii_phy`

System Flow Capability

Node name	`$bsp_path/config/system_flow_capability`
Description	Indicates system flow capability for thermal management. Set to "C2P" (Cold to Power) for systems that support this flow direction.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/system_flow_capability C2P`

Fan Direction EEPROM

Node name	`$bsp_path/config/fan_dir_eeprom`
Description	Enables fan direction detection from EEPROM. Set to "1" for systems that support fan direction detection via EEPROM. On these systems, each fan drawer direction (`thermal/fan<index>_dir`) is taken from the fan module EEPROM VPD. When the fan is removed, the corresponding `fan<index>_dir` attribute may also be removed. See Read Fan Direction.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/fan_dir_eeprom 1`

Global Write Protection Wait Step

Node name	`$bsp_path/config/global_wp_wait_step`
Description	Write protection wait step configuration in seconds. Used by global write protection mechanism.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/global_wp_wait_step 1`

Global Write Protection Timeout

Node name	`$bsp_path/config/global_wp_timeout`
Description	Write protection timeout configuration in seconds. Used by global write protection mechanism.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/global_wp_timeout 20`

ConnectX Default I2C Bus

Node name	`$bsp_path/config/cx_default_i2c_bus`
Description	Default I2C bus number for ConnectX devices. Used for ConnectX device communication.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/cx_default_i2c_bus 15`

JTAG Bridge Offset

Node name	`$bsp_path/config/jtag_bridge_offset`
Description	JTAG bridge memory offset for debugging purposes. Extracted from /proc/iomem mlxplat_jtag_bridge entry.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/jtag_bridge_offset 0x40000000`

Core 0 Temperature ID

Node name	`$bsp_path/config/core0_temp_id`
Description	Temperature sensor ID for CPU core 0. Used for thermal monitoring of the first CPU core.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/core0_temp_id 2`

Core 1 Temperature ID

Node name	`$bsp_path/config/core1_temp_id`
Description	Temperature sensor ID for CPU core 1. Used for thermal monitoring of the second CPU core.
Access	Read only
Release version	3.0
Example	`cat $bsp_path/config/core1_temp_id 3`

Read PDB Hotswap Scale Factor

Stack	Host
Node name	`$bsp_path/config/pdb_hotswap_scale`
Description	LM5066I PDB hot-swap input power and current scaling factor. Written by `sn66xxld_specific()` in `hw-management.sh` for SN6600_LD (SKU HI193). The same value is symlinked under each lm5066i PDB hotswap environment node as `*_power1_scale` and `*_curr1_scale` (see Environment Control). lm-sensors applies the same factor via `compute` rules in `usr/etc/hw-management-sensors/sn66xxld_sensors.conf`.
Values	Scale, Float: **5.333** on SN6600_LD (validated in `hw-management.sh`)
Access	Read only
Release version	V.7.0070.1000
Example	`cat $bsp_path/config/pdb_hotswap_scale`

Read LED Control Type Map

Stack	Host
Node name	`$bsp_path/config/led_control_type`
Description	Optional platform map of LED name (or glob mask) to control owner. Written by `set_config_data()` in `hw-management.sh` when a `*_specific()` function sets the `led_control_type` array. Space-separated pairs: `name type [name type ...]`. If the node is absent, or a given LED name is not listed, LED add uses the default owner `led_hw_sw` (see Get LED Control Owner). Name matching (in `hw-management-chassis-events.sh get_led_control_type()`): 1. Exact match of the udev LED name (`status`, `fan`, `fan1`) or `led_<name>`. `fan` and `fan1` are different names. 2. Glob mask: `*` matches any string, `?` matches one character. First matching mask wins. Masks must be quoted in the platform array (`"fan*"`, `"led?"`) so the shell does not expand them. 3. Otherwise `led_hw_sw`.
Values	name, String: LED name (`fan`, `status`) or mask (`fan*`, `led_psu?`, `led*`); type, String: `led_sw`, `led_hw`, `led_hw_sw`
Access	Read only
Release version	3.2.7
Example	`cat $bsp_path/config/led_control_type # fan led_sw psu led_sw status led_sw`

Cable Cartridge FRU Validity

Stack	Host
Node name	`$bsp_path/config/cable_cartridge<cartridge number>_valid`
Description	Result of the IPMI FRU check run on the cable cartridge EEPROM at init. The BMC reads the same EEPROM and programs the rack id, topology id, switch tray id and slot id into the switch board CPLD without validating it, so this node reports whether that source data can be trusted. Written by `validate_cartridge_fru()` in `hw-management-chassis-events.sh` after `ipmi-fru` parses `eeprom/cable_cartridge<n>_eeprom`. `ipmi-fru` exits 0 even for a broken FRU, so the verdict comes from its output: a `FRU Error` line, or a missing board serial number or chassis custom info field, gives 0. Failures are also reported to syslog. Created on the N51XX_LD and N61XX_LD families only. Other platforms that have cable cartridge EEPROMs parse them through a different path and do not get this node. It is also absent when `ipmi-fru` is not installed. The node is created when the cartridge EEPROM appears and removed when the cartridge is removed, so it never reports a verdict for an empty slot. One node per cartridge, so the count follows `config/cartridge_counter`, which is platform dependent: 4 on N6100_LD (HI180), N6300_LD (HI185), N5110_LD (HI166) and N5100_LD (HI167/HI170), and 2 on N5112_LD (HI169).
Values	Validity, Integer: 1 - FRU valid, 0 - FRU invalid
Access	Read only
Release version	3.2.8
Example	Check cable cartridge 1 FRU validity: `cat $bsp_path/config/cable_cartridge1_valid`

BIOS Control

BIOS Active Image

Node name	`$bsp_path/system/bios_active_image`
Description	Currently active BIOS image identifier. Indicates which BIOS image is currently running on the system.
Values	Text: Image identifier
Access	Read only
Release version	3.0
Example	`cat $bsp_path/system/bios_active_image primary`

EEPROM Control

Read CPU EEPROM Data

Node name	`$bsp_path/eeprom/cpu_info`
Description	Read CPU raw data in hexadecimal format
Values	EEPROM information, Hex: Hex dump format of memory
Access	Read only
Release version	1.0
Example	Get CPU EEPROM information: `cat $bsp_path/eeprom/cpu_info`

Read Fan Module EEPROM Data

Node name	`$bsp_path/eeprom/fan<fan module number>_info`
Description	Read fan module raw data in hexadecimal format Note: This attribute is not supported on Comex CPU systems.
Values	EEPROM information, Hex: Hex dump format of memory
Access	Read only
Release version	1.0
Example	Get fan module 1 EEPROM information: `hexdump -C $bsp_path/eeprom/fan1_info`

Read Power Supply Module EEPROM Data

Node name	`$bsp_path/eeprom/psu<power supply module number>_info`
Description	Read power supply module raw data in hexadecimal format
Values	EEPROM information, Hex: Hex dump format of memory
Access	Read only
Release version	1.0
Example	Get power supply module 1 EEPROM information: `cat $bsp_path/eeprom/psu1_info`

Read System Chassis EEPROM Data

Node name	`$bsp_path/eeprom/vpd_info`
Description	Read system chassis raw data in hexadecimal format
Values	EEPROM information, Hex: Hex dump format of memory
Access	Read only
Release version	1.0
Example	Get system chassis EEPROM information: `cat $bsp_path/eeprom/vpd_info`

Read System Chassis EEPROM Parsed Data

Node name	`$bsp_path/eeprom/vpd_data`
Description	Read system chassis parsed data in text format
Values	EEPROM information, string: string format of memory
Access	Read only
Release version	1.0
Example	Get system chassis EEPROM information: `cat $bsp_path/eeprom/vpd_data`

Read Cable Cartridge EEPROM Data

Node name	`$bsp_path/eeprom/cable_cartridge<index>_eeprom` `$bsp_path/eeprom/cable_cartridge<index>_eeprom_data`
Description	Read cable cartridge EEPROM raw data or parsed data. Cable cartridges are used in liquid-cooled multi-ASIC systems for connecting external cables. Note: This attribute is for systems with cable cartridges (N61XX_LD family: N6100_LD, N6300_LD). N6100_LD has 4 cable cartridges (index 1-4) on I2C buses 68, 69, 70, 71. N6300_LD (SKU HI185) uses the same cartridge EEPROM connect table as N6100_LD (`n61xxld_cartridge_eeprom_connect_table` in `hw-management.sh`).
Values	index, Integer: 1-4 (N6100_LD); EEPROM data, Binary/String: Raw or parsed EEPROM content
Access	Read only
Release version	V.7.0060.1000
Example	Get cable cartridge EEPROM data: `cat $bsp_path/eeprom/cable_cartridge1_eeprom_data cat $bsp_path/eeprom/cable_cartridge2_eeprom_data cat $bsp_path/eeprom/cable_cartridge3_eeprom_data cat $bsp_path/eeprom/cable_cartridge4_eeprom_data`

Read system EEPROM (BMC stack)

Stack	BMC (`hw-management-bmc-events.sh`, event `eeprom_system`)
Node name	`$bsp_path/eeprom/eeprom_system`
Description	Raw system VPD EEPROM on the **BMC** image. HI189 udev matches I2C `5-0051` (`hw-management-bmc-early-i2c-devices.json`: bus 5 / `0x51`, **24c512**). Symlink points at the kernel `eeprom` attribute for that device.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/eeprom/eeprom_system`

Read BMC board EEPROM (BMC stack)

Stack	BMC (`hw-management-bmc-events.sh`, event `eeprom_bmc`)
Node name	`$bsp_path/eeprom/eeprom_bmc`
Description	BMC FRU EEPROM on the **BMC** image. HI189 udev matches I2C `4-0050` (bus 4 / `0x50`, **24c02** per early-I2C JSON).
Access	Read only
Release version	1.0
Example	`cat $bsp_path/eeprom/eeprom_bmc` **Reference:** `bmc/examples/hw-management-bmc-eeprom-sysfs.txt`

Environment Control

Get A2D Voltage

Node name	`$bsp_path/environment/a2d_iio:device<number>_raw<index>`
Description	Get raw voltage input from A2D sensor
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get voltage input from A2D1: `cat $bsp_path/environment/a2d_iio:device0_raw_1`

Get Comex Voltage Current

Node name	`$bsp_path/environment/comex_voltmon<index>_curr<index>_input`
Description	Get raw voltage input from Comex Note: This attribute is for Comex based system only
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get comex voltage monitor 1 current2 reading: `cat $bsp_path/environment/comex_voltmon1_curr2_input`

Get Comex Voltage Input

Node name	`$bsp_path/environment/comex_voltmon<index>_in<index>_input`
Description	Get raw voltage input from Comex Note: This attribute is for Comex based system only
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get Comex voltage monitor 1 input reading: `cat $bsp_path/environment/comex_voltmon1_in1_input`

Get Comex Voltage Power

Node name	`$bsp_path/environment/comex_voltmon<index>_power<index>_input`
Description	Get raw voltage input from Comex Note: This attribute is for Comex based system only
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get Comex voltage monitor 1 power reading: `cat $bsp_path/environment/comex_power2_input`

Get System Voltage Current

Node name	`$bsp_path/environment/voltmon<index>_curr<index>_input`
Description	Get raw voltage input from system
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get voltage monitor 1 current2 reading: `cat $bsp_path/environment/voltmon1_curr2_input`

Get System Voltage Input

Node name	`$bsp_path/environment/voltmon<index>_in<index>_input`
Description	Get raw voltage input from system
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get voltage monitor 1 input reading: `cat $bsp_path/environment/voltmon1_in1_input`

Get System Voltage Power

Node name	`$bsp_path/environment/voltmon<index>_power<index>_input`
Description	Get raw voltage input from system
Values	Voltage, Integer: X
Access	Read only
Release version	1.0
Example	Get voltage monitor 1 power reading: `cat $bsp_path/environment/voltmon1_power2_input`

Get PDB Hotswap Controller Current

Node name	`$bsp_path/environment/pdb_hotswap<index>_curr<index>_input`
Description	Get PDB (Power Distribution Board) hot-swap controller current measurement Note: This attribute is for liquid-cooled systems only (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD; SN66XX_LD family: SN6600_LD)
Values	Current, Integer: Value in milliamps (mA)
Access	Read only
Release version	V.7.0050.3000
Example	Get PDB hotswap 1 current input: `cat $bsp_path/environment/pdb_hotswap1_curr1_input`

Get PDB Hotswap Controller Voltage

Node name	`$bsp_path/environment/pdb_hotswap<index>_in<index>_input`
Description	Get PDB hot-swap controller voltage measurement Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Voltage, Integer: Value in millivolts (mV)
Access	Read only
Release version	V.7.0050.3000
Example	Get PDB hotswap 1 input voltage: `cat $bsp_path/environment/pdb_hotswap1_in1_input`

Get PDB Hotswap Controller Power

Node name	`$bsp_path/environment/pdb_hotswap<index>_power<index>_input`
Description	Get PDB hot-swap controller power measurement Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Power, Integer: Value in microwatts
Access	Read only
Release version	V.7.0050.3000
Example	Get PDB hotswap 1 power input: `cat $bsp_path/environment/pdb_hotswap1_power1_input` Note: On SN6600_LD, when the underlying hwmon device is **lm5066i**, `hw-management-chassis-events.sh` creates scale-factor symlinks for input power and current (see following sections). lm-sensors also applies factor **5.333** to the lm5066i PDB hotswap `power1` and `curr1` labels in `sn66xxld_sensors.conf`.

Get PDB Hotswap Power Scale

Stack	Host
Node name	`$bsp_path/environment/pdb_hotswap<index>_power1_scale`
Description	Scaling factor symlink for PDB hot-swap controller input power. Created when the hotswap hwmon driver name is **lm5066i**; the symlink targets `$bsp_path/config/pdb_hotswap_scale`. Note: SN6600_LD (SKU HI193) only. Not created for other hotswap driver types (for example mp5926).
Access	Read only
Release version	V.7.0070.1000
Example	`cat $bsp_path/environment/pdb_hotswap1_power1_scale`

Get PDB Hotswap Current Scale

Stack	Host
Node name	`$bsp_path/environment/pdb_hotswap<index>_curr1_scale`
Description	Scaling factor symlink for PDB hot-swap controller input current. Created when the hotswap hwmon driver name is **lm5066i**; the symlink targets `$bsp_path/config/pdb_hotswap_scale`. Note: SN6600_LD (SKU HI193) only.
Access	Read only
Release version	V.7.0070.1000
Example	`cat $bsp_path/environment/pdb_hotswap1_curr1_scale`

Get PDB Hotswap Controller Thresholds

Node name	`$bsp_path/environment/pdb_hotswap<index>_<sensor>_<threshold>`
Description	Get PDB hot-swap controller threshold values (crit, lcrit, max, min) Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Threshold, Integer: Varies by sensor type
Access	Read only
Release version	V.7.0050.3000
Example	Get PDB hotswap 1 current max threshold: `cat $bsp_path/environment/pdb_hotswap1_curr1_max cat $bsp_path/environment/pdb_hotswap1_in1_crit cat $bsp_path/environment/pdb_hotswap1_in1_lcrit`

Get PDB Power Converter Current

Node name	`$bsp_path/environment/pdb_pwr_conv<index>_curr<index>_input` (SN58XX_LD, SN66XX_LD) `$bsp_path/environment/pwr_conv<index>_curr<index>_input` (N61XX_LD)
Description	Get PDB power converter current measurement (input or output) Note: This attribute is for liquid-cooled systems only. - SN58XX_LD family uses: `pdb_pwr_conv<N>_...` naming (1 power converter on SN5810_LD; more on SN5800_LD) - SN66XX_LD family (SN6600_LD) uses: `pdb_pwr_conv<N>_...` with N = 1..2 - N61XX_LD family uses: `pwr_conv<N>_...` naming (2 power converters)
Values	Current, Integer: Value in milliamps (mA)
Access	Read only
Release version	V.7.0050.3000 (SN58XX_LD), V.7.0060.1000 (N61XX_LD)
Example	Get power converter current readings: `# SN58XX_LD systems cat $bsp_path/environment/pdb_pwr_conv1_curr1_input cat $bsp_path/environment/pdb_pwr_conv1_curr2_input # SN6600_LD systems (2 PDB converters) cat $bsp_path/environment/pdb_pwr_conv1_curr1_input cat $bsp_path/environment/pdb_pwr_conv1_curr2_input cat $bsp_path/environment/pdb_pwr_conv2_curr1_input cat $bsp_path/environment/pdb_pwr_conv2_curr2_input # N6100_LD systems (2 power converters) cat $bsp_path/environment/pwr_conv1_curr1_input cat $bsp_path/environment/pwr_conv1_curr2_input cat $bsp_path/environment/pwr_conv2_curr1_input cat $bsp_path/environment/pwr_conv2_curr2_input`

Get PDB Power Converter Voltage

Node name	`$bsp_path/environment/pdb_pwr_conv<index>_in<index>_input` (SN58XX_LD, SN66XX_LD) `$bsp_path/environment/pwr_conv<index>_in<index>_input` (N61XX_LD)
Description	Get PDB power converter voltage measurement (input or output) Note: This attribute is for liquid-cooled systems only. - SN58XX_LD and SN66XX_LD families use: `pdb_pwr_conv<N>_...` naming - N61XX_LD family uses: `pwr_conv<N>_...` naming
Values	Voltage, Integer: Value in millivolts (mV)
Access	Read only
Release version	V.7.0050.3000 (SN58XX_LD), V.7.0060.1000 (N61XX_LD)
Example	Get power converter voltage readings: `# SN58XX_LD systems cat $bsp_path/environment/pdb_pwr_conv1_in1_input cat $bsp_path/environment/pdb_pwr_conv1_in2_input # SN6600_LD systems cat $bsp_path/environment/pdb_pwr_conv1_in1_input cat $bsp_path/environment/pdb_pwr_conv1_in2_input cat $bsp_path/environment/pdb_pwr_conv2_in1_input cat $bsp_path/environment/pdb_pwr_conv2_in2_input # N6100_LD systems cat $bsp_path/environment/pwr_conv1_in1_input cat $bsp_path/environment/pwr_conv1_in2_input cat $bsp_path/environment/pwr_conv2_in1_input cat $bsp_path/environment/pwr_conv2_in2_input`

Get PDB Power Converter Power

Node name	`$bsp_path/environment/pdb_pwr_conv<index>_power<index>_input` (SN58XX_LD, SN66XX_LD) `$bsp_path/environment/pwr_conv<index>_power<index>_input` (N61XX_LD)
Description	Get PDB power converter power measurement (input or output) Note: This attribute is for liquid-cooled systems only. - SN58XX_LD and SN66XX_LD families use: `pdb_pwr_conv<N>_...` naming - N61XX_LD family uses: `pwr_conv<N>_...` naming
Values	Power, Integer: Value in microwatts
Access	Read only
Release version	V.7.0050.3000 (SN58XX_LD), V.7.0060.1000 (N61XX_LD)
Example	Get power converter power readings: `# SN58XX_LD systems cat $bsp_path/environment/pdb_pwr_conv1_power1_input # SN6600_LD systems cat $bsp_path/environment/pdb_pwr_conv1_power1_input cat $bsp_path/environment/pdb_pwr_conv2_power1_input # N6100_LD systems cat $bsp_path/environment/pwr_conv1_power1_input cat $bsp_path/environment/pwr_conv2_power1_input`

Get PDB Power Converter Thresholds

Node name	`$bsp_path/environment/pdb_pwr_conv<index>_<sensor>_<threshold>`
Description	Get PDB power converter threshold values (crit, lcrit, max, min) Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Threshold, Integer: Varies by sensor type
Access	Read only
Release version	V.7.0050.3000
Example	Get PDB power converter 1 thresholds: `cat $bsp_path/environment/pdb_pwr_conv1_curr1_crit cat $bsp_path/environment/pdb_pwr_conv1_in1_max cat $bsp_path/environment/pdb_pwr_conv1_in1_min`

Events

Get FAN hot-plug event status

Node name	`$bsp_path/events/fan<index>`
Description	Get hot-plug event status of FAN<index> Index max value corresponds to $bsp_path/config/hotplug_fans 0 – FAN<index> was removed, 1 – FAN<index> was inserted.
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1300
Example	Get FAN3 hot-plug status: `cat $bsp_path/events/fan3`

Get PSU hot-plug event status

Node name	`$bsp_path/events/psu<index>`
Description	Get hot-plug event status of PSU<index> Index max value corresponds to $bsp_path/config/hotplug_psus 0 – PSU<index> was removed, 1 – PSU<index> was inserted.
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1300
Example	Get PSU2 hot-plug status: `cat $bsp_path/events/psu2`

PWR hot-plug event status

Node name	`$bsp_path/events/pwr<index>`
Description	Get latest hot-plug event status of PWR<index> Index max value corresponds to $bsp_path/config/hotplug_pwrs 0 – PWR<index> cable was plugged-out, 1 – PWR<index> cable was plugged-in.
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1300
Example	Get Power1 cable hot-plug status: `cat $bsp_path/events/pwr1`

PDB hot-plug event status

Node name	`$bsp_path/events/pdb<index>`
Description	Get hot-plug event status of PDB (Power Distribution Board) <index> Index max value corresponds to $bsp_path/config/hotplug_pdbs 0 – PDB<index> was removed, 1 – PDB<index> was inserted. Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Event, Integer: 0 / 1
Access	Read
Release version	V.7.0050.3000
Example	Get PDB1 hot-plug status: `cat $bsp_path/events/pdb1`

DPU Ready event

Node name	`$bsp_path/events/dpu[1-8]_ready`
Description	Get dpu ready event
Values	dpu, Integer: 0 / 1
Access	Read
Release version	V.7.0030.4000
Example	Get dpu[1-8] ready status `cat $bsp_path/events/dpu1_ready`

DPU Shutdown event

Node name	`$bsp_path/events/dpu[1-8]_shtdn_ready`
Description	Get dpu shutdown ready event
Values	dpu, Integer: 0 / 1
Access	Read
Release version	V.7.0030.4000
Example	Get dpu[1-8] shut down ready status `cat $bsp_path/events/dpu1_shtdn_ready`

EROT (External Root of Trust) AP event

Node name	`$bsp_path/events/erot<index>_ap`
Description	EROT Application Processor event status. Indicates the state of the External Root of Trust application processor at the specified index. Note: This attribute is for systems with eRoT support (N61XX_LD family: N6100_LD). N6100_LD has 1 eRoT (index=1).
Values	index, Integer: 1 (N6100_LD); Status, Integer: 0 (inactive), 1 (active)
Access	Read only
Release version	V.7.0060.1000
Example	Get eRoT AP status: `cat $bsp_path/events/erot1_ap`

EROT (External Root of Trust) Error Event

Node name	`$bsp_path/events/erot<index>_error`
Description	EROT error event status. Indicates error conditions detected by the External Root of Trust at the specified index. Note: This attribute is for systems with eRoT support (N61XX_LD family: N6100_LD). N6100_LD has 1 eRoT (index=1).
Values	index, Integer: 1 (N6100_LD); Status, Integer: 0 (no error), 1 (error detected)
Access	Read only
Release version	V.7.0060.1000
Example	Get eRoT error status: `cat $bsp_path/events/erot1_error`

Graceful Power Off Event

Node name	`$bsp_path/events/graceful_pwr_off`
Description	Graceful power off event status. Indicates when the system is performing a graceful shutdown sequence.
Values	Integer: 0 (normal), 1 (graceful shutdown)
Access	Read only
Release version	3.0
Example	Read the node: `cat $bsp_path/events/graceful_pwr_off`

Power Button Event (events)

Node name	`$bsp_path/events/power_button`
Description	Power button event status. Indicates when the system power button has been pressed.
Values	Integer: 0 (not pressed), 1 (pressed)
Access	Read only
Release version	3.0
Example	Read the node: `cat $bsp_path/events/power_button`

Power Events

Node name	`$bsp_path/config/power_events`
Description	Power events configuration file. Contains settings and thresholds for power-related event handling.
Values	Text file: Configuration data
Access	Read only
Release version	3.0
Example	Read the node: `cat $bsp_path/config/power_events`

Power Button Event (system)

Node name	`$bsp_path/system/power_button_evt`
Description	Power button event status. Indicates the current state of power button events in the system.
Values	Integer: Event status
Access	Read only
Release version	3.0
Example	`cat $bsp_path/system/power_button_evt`

Alarms

Get PMBUS voltmon alarm status

Node name	`$bsp_path/alarm/voltmon<index>_<sensor_name>_alarm`
Description	Get voltmonitor alarm status of voltmon<index> sensor_name one of: in, curr, power, temp. sensor count can be different for different voltmonitor types/configuration. Alarm set by voltmon sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1000
Example	Get voltmon1_in1_alarm: `cat $bsp_path/alarm/voltmon1_in1_alarm`

Get COMEX PMBUS voltmon alarm status

Node name	`$bsp_path/alarm/comex_voltmon<index>_<sensor_name>_alarm`
Description	Get voltmonitor alarm status of comex_voltmon<index> sensor_name one of: in, curr, power, temp. sensor count can be different for different voltmonitor types/configuration. Alarm set by voltmon sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1000
Example	Get comex_voltmon1_in1_alarm: `cat $bsp_path/alarm/comex_voltmon1_in1_alarm`

Get PSU PMBUS alarm status

Node name	`$bsp_path/alarm/psu<index>_<sensor_name>_alarm`
Description	Get PSU PMBUS alarm status of PSU<index> sensor_name: in, curr, power, temp. sensor count can be different for different PSU types/configuration. Alarm set by PSU sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1000
Example	Get psu1_power1_alarm: `cat $bsp_path/alarm/psu1_power1_alarm`

Get PDB Hotswap Controller alarm status

Node name	`$bsp_path/alarm/pdb_hotswap<index>_<sensor_name>_alarm`
Description	Get PDB (Power Distribution Board) hot-swap controller alarm status sensor_name: in, curr, power, temp. Alarm set by PDB hotswap controller sensor itself (hardware-controlled attribute) 1 – alarm set 0 – alarm clear Note: This attribute is for liquid-cooled systems only (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD; SN66XX_LD family: SN6600_LD)
Values	Alarm, Integer: 0 / 1
Access	Read
Release version	V.7.0050.3000
Example	Get PDB hotswap 1 alarms: `cat $bsp_path/alarm/pdb_hotswap1_curr1_alarm cat $bsp_path/alarm/pdb_hotswap1_in1_alarm cat $bsp_path/alarm/pdb_hotswap1_power1_alarm cat $bsp_path/alarm/pdb_hotswap1_temp1_crit_alarm cat $bsp_path/alarm/pdb_hotswap1_temp1_max_alarm`

Get PDB Power Converter alarm status

Node name	`$bsp_path/alarm/pdb_pwr_conv<index>_<sensor_name>_alarm`
Description	Get PDB power converter alarm status sensor_name: in, curr, power, temp. Alarm set by PDB power converter sensor itself (hardware-controlled attribute) 1 – alarm set 0 – alarm clear Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Alarm, Integer: 0 / 1
Access	Read
Release version	V.7.0050.3000
Example	Get PDB power converter 1 alarms: `cat $bsp_path/alarm/pdb_pwr_conv1_curr1_alarm cat $bsp_path/alarm/pdb_pwr_conv1_curr2_alarm cat $bsp_path/alarm/pdb_pwr_conv1_in1_alarm cat $bsp_path/alarm/pdb_pwr_conv1_in2_alarm cat $bsp_path/alarm/pdb_pwr_conv1_power1_alarm cat $bsp_path/alarm/pdb_pwr_conv1_temp1_crit_alarm cat $bsp_path/alarm/pdb_pwr_conv1_temp1_max_alarm`

Get CPU temp alarm status

Node name	`$bsp_path/alarm/cpu_core<idx>_crit_alarm`
Description	Get CPU core overtemperature alarm status. idx – 0...cpu_core_num cpu_core_num can be different for different CPU type Alarm set by CPU sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear
Values	Thermal, Integer: 0 / 1
Access	Read
Release version	V.7.0010.1000
Example	Get cpu_core0_crit_alarm: `cat $bsp_path/alarm/cpu_core0_crit_alarm`

PSU FW

Get Secondary FW version of PSU

Node name	`$bsp_path/firmware/psu<index>_fw_ver`
Description	Get secondary FW version of PSU<index> For Murata 1500/2000 and Delta 550 the contents of the file is the relevant FW version For all other PSUs - the contents is string ”N/A”
Values	version, string
Access	Read
Release version	V.7.0020.2000
Example	Get secondary FW version of PSU1 `$bsp_path/firmware/psu1_fw_ver`

Get Primary FW version of PSU

Node name	`$bsp_path/firmware/psu<index>_fw_primary_ver`
Description	Get primary FW version of PSU<index> Primary files exist only for Murata. For all other PSUs - the contents is string ”N/A”
Values	version, string
Access	Read
Release version	V.7.0020.2000
Example	Get primary FW version of PSU1 `$bsp_path/firmware/psu1_fw_primary_ver`

DPU system attributes

Get DPU id

Node name	`$bsp_path/system/dpu<index>_id`
Description	Get DPU<index> id
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` id `$bsp_path/system/dpu1_id`

Get DPU boot progress

Node name	`$bsp_path/system/dpu<index>_boot_progress`
Description	Get DPU<index> boot progress
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` boot progress `$bsp_path/system/dpu1_boot_progress`

Get DPU cpld version

Node name	`$bsp_path/system/dpu<index>_cpld_version`
Description	Get DPU<index> CPLD version
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 cpld version `$bsp_path/system/dpu1_cpld_version`

Get DPU cpld base version

Node name	`$bsp_path/system/dpu<index>_cpld_base_version`
Description	Get DPU<index> CPLD base version
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 cpld base version `$bsp_path/system/dpu1_cpld_base_version`

Get DPU auxiliary power reset reason

Node name	`$bsp_path/system/dpu<index>_aux_pwr_rst_reason`
Description	Get DPU<index> auxiliary power reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 auxiliary power reset reason `$bsp_path/system/dpu1_aux_pwr_rst_reason`

Get DPU main board reset reason

Node name	`$bsp_path/system/dpu<index>_main_brd_rst_reason`
Description	Get DPU<index> main board reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` main board reset reason `$bsp_path/system/dpu1_main_brd_rst_reason`

Get DPU comex power failure

Node name	`$bsp_path/system/dpu<index>_comex_pwr_fail`
Description	Get DPU<index> comex power failure
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` comex power failure `$bsp_path/system/dpu1_comex_pwr_fail`

Get DPU power reset reason

Node name	`$bsp_path/system/dpu<index>_pwr_rst_reason`
Description	Get DPU<index> power reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` power reset reason `$bsp_path/system/dpu1_pwr_rst_reason`

Get DPU thermal shutdown reason

Node name	`$bsp_path/system/dpu<index>_thermal_shtdn_reason`
Description	Get DPU<index> thermal shutdown reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` thermal shutdown reason `$bsp_path/system/dpu1_thermal_shtdn_reason`

Get DPU tpm reset reason

Node name	`$bsp_path/system/dpu<index>_tpm_rst_reason`
Description	Get DPU<index> tpm reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` tpm reset reason `$bsp_path/system/dpu1_tpm_rst_reason`

Get DPU perst reset reason

Node name	`$bsp_path/system/dpu<index>_perst_rst_reason`
Description	Get DPU<index> perst reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` perst reset reason `$bsp_path/system/dpu1_perst_rst_reason`

Get DPU phy reset reason

Node name	`$bsp_path/system/dpu<index>_phy_rst_reason`
Description	Get DPU<index> phy reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` phy reset reason `$bsp_path/system/dpu1_phy_rst_reason`

Get DPU usb phy reset reason

Node name	`$bsp_path/system/dpu<index>_usb_phy_rst_reason`
Description	Get DPU<index> usb phy reset reason
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` usb phy reset reason `$bsp_path/system/dpu1_usb_phy_rst_reason`

Get DPU fpga part number

Node name	`$bsp_path/system/dpu<index>_fpga_part_number`
Description	Get DPU<index> fpga part number
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` fpga part number `$bsp_path/system/dpu1_fpga_part_number`

Get DPU fpga minor version

Node name	`$bsp_path/system/dpu<index>_fpga_minor_version`
Description	Get DPU<index> fpga minor version
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` fpga minor version `$bsp_path/system/dpu1_fpga_minor_version`

Get DPU ufm upgrade status

Node name	`$bsp_path/system/dpu<index>_ufm_upgrade_status`
Description	Get DPU<index> ufm upgrade status
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` ufm upgrade status `$bsp_path/system/dpu1_ufm_upgrade_status`

Get DPU VR update status

Node name	`$bsp_path/system/dpu<index>_vr_update_status`
Description	Get DPU<index> VR update status
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` VR update status `$bsp_path/system/dpu1_vr_update_status`

DPU events

Get DPU PLL power good indication

Node name	`$bsp_path/events/dpu<index>_pll_pwr_good`
Description	Get DPU<index> PLL power good indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 PLL power good indication `$bsp_path/events/dpu1_pll_pwr_good`

Get DPU input power indication

Node name	`$bsp_path/events/dpu<index>_input_pwr`
Description	Get DPU<index> input power indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 input power indication `$bsp_path/events/dpu1_input_pwr`

Get DPU serdes power indication

Node name	`$bsp_path/events/dpu<index>_serdes_pwr`
Description	Get DPU<index> serdes power indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` serdes power indication `$bsp_path/events/dpu1_serdes_pwr`

Get DPU serdes analog power indication

Node name	`$bsp_path/events/dpu<index>_serdes_analog_pwr`
Description	Get DPU<index> serdes analog power indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` serdes analog power indication `$bsp_path/events/dpu1_serdes_analog_pwr`

Get DPU core power indication

Node name	`$bsp_path/events/dpu<index>_core_pwr`
Description	Get DPU<index> core power indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` core power indication `$bsp_path/events/dpu1_core_pwr`

Get DPU cpu power indication

Node name	`$bsp_path/events/dpu<index>_cpu_pwr`
Description	Get DPU<index> cpu power indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` cpu power indication `$bsp_path/events/dpu1_cpu_pwr`

Get DPU digital interfaces power

Node name	`$bsp_path/events/dpu<index>_digital_if_pwr`
Description	Get DPU<index> digital interfaces power
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` digital interfaces power `$bsp_path/events/dpu1_digital_if_pwr`

Get DPU ddr5 power indication

Node name	`$bsp_path/events/dpu<index>_ddr5_pwr`
Description	Get DPU<index> ddr5 power indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` ddr5 power indication `$bsp_path/events/dpu1_ddr5_pwr`

Get DPU thermal trip indication

Node name	`$bsp_path/events/dpu<index>_thermal_trip`
Description	Get DPU<index> thermal trip indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` thermal trip indication `$bsp_path/events/dpu1_thermal_trip`

Get DPU tps upgrade status

Node name	`$bsp_path/events/dpu<index>_tps_upgrade`
Description	Get DPU<index> tps upgrade status
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` tps upgrade status `$bsp_path/events/dpu1_tps_upgrade`

Get DPU cpu power fault indication

Node name	`$bsp_path/events/dpu<index>_cpu_pwr_fault`
Description	Get DPU<index> cpu power fault indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get `dpu1` cpu power fault indication `$bsp_path/events/dpu1_cpu_pwr_fault`

Get DPU cpu VR hot alert

Node name	`$bsp_path/events/dpu<index>_cpu_vr_hot`
Description	Get DPU<index> cpu VR hot alert
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 cpu VR hot alert `$bsp_path/events/dpu1_cpu_vr_hot`

Get DPU ddr5 fault indication

Node name	`$bsp_path/events/dpu<index>_ddr5_fault`
Description	Get DPU<index> ddr5 fault indication
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 ddr5 fault indication `$bsp_path/events/dpu1_ddr5_fault`

Get DPU ddr5 hot alert

Node name	`$bsp_path/events/dpu<index>_ddr5_hot`
Description	Get DPU<index> ddr5 hot alert
Values	dpu, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get dpu1 ddr5 hot alert `$bsp_path/events/dpu1_ddr5_hot`

LC Alarms

Get LC Hot Swap Power Alarm

Node name	`$bsp_path/alarm/lc<index>_hotswap_pwr_alarm`
Description	Get LC<index> hot swap power alarm
Values	lc, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get lc1 hot swap power alarm `$bsp_path/alarm/lc1_hotswap_pwr_alarm`

Get LC Voltage Input Alarm

Node name	`$bsp_path/alarm/lc<index>_volt_in_alarm`
Description	Get LC<index> voltage input alarm
Values	lc, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get lc1 voltage input alarm `$bsp_path/alarm/lc1_volt_in_alarm`

Get LC Voltage Current Alarm

Node name	`$bsp_path/alarm/lc<index>_volt_curr_alarm`
Description	Get LC<index> voltage current alarm
Values	lc, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get lc1 voltage current alarm `$bsp_path/alarm/lc1_volt_curr_alarm`

Get LC Voltage Power Alarm

Node name	`$bsp_path/alarm/lc<index>_volt_pwr_alarm`
Description	Get LC<index> voltage power alarm
Values	lc, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get lc1 voltage power alarm `$bsp_path/alarm/lc1_volt_pwr_alarm`

LC EEPROM

Read LC EEPROM FRU

Node name	`$bsp_path/eeprom/lc<index>_fru`
Description	Read LC<index> EEPROM FRU data in hexadecimal format
Values	EEPROM information, Hex: Hex dump format of memory
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 EEPROM FRU information: `cat $bsp_path/eeprom/lc1_fru`

Read LC EEPROM INI

Node name	`$bsp_path/eeprom/lc<index>_ini`
Description	Read LC<index> EEPROM INI data in hexadecimal format
Values	EEPROM information, Hex: Hex dump format of memory
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 EEPROM INI information: `cat $bsp_path/eeprom/lc1_ini`

Read LC EEPROM VPD Parsed

Node name	`$bsp_path/eeprom/lc<index>_vpd_parsed`
Description	Read LC<index> EEPROM VPD parsed data in text format
Values	EEPROM information, string: Parsed VPD data
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 EEPROM VPD parsed information: `cat $bsp_path/eeprom/lc1_vpd_parsed`

Read LC EEPROM INI Parsed

Node name	`$bsp_path/eeprom/lc<index>_ini_parsed`
Description	Read LC<index> EEPROM INI parsed data in text format
Values	EEPROM information, string: Parsed INI data
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 EEPROM INI parsed information: `cat $bsp_path/eeprom/lc1_ini_parsed`

LC Environment

Get LC Voltage Current

Node name	`$bsp_path/environment/lc<index>_volt_curr<idx>_input`
Description	Get LC<index> voltage current reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 voltage current reading: `cat $bsp_path/environment/lc1_volt_curr1_input`

Get LC Voltage Input

Node name	`$bsp_path/environment/lc<index>_volt_in<idx>_input`
Description	Get LC<index> voltage input reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 voltage input reading: `cat $bsp_path/environment/lc1_volt_in1_input`

Get LC Voltage Power

Node name	`$bsp_path/environment/lc<index>_volt_pwr<idx>_input`
Description	Get LC<index> voltage power reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 voltage power reading: `cat $bsp_path/environment/lc1_volt_pwr1_input`

Get LC Hot Swap Current

Node name	`$bsp_path/environment/lc<index>_hotswap_curr<idx>_input`
Description	Get LC<index> hot swap current reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 hot swap current reading: `cat $bsp_path/environment/lc1_hotswap_curr1_input`

Get LC Hot Swap Input

Node name	`$bsp_path/environment/lc<index>_hotswap_in<idx>_input`
Description	Get LC<index> hot swap input reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 hot swap input reading: `cat $bsp_path/environment/lc1_hotswap_in1_input`

Get LC Hot Swap Power

Node name	`$bsp_path/environment/lc<index>_hotswap_pwr<idx>_input`
Description	Get LC<index> hot swap power reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 hot swap power reading: `cat $bsp_path/environment/lc1_hotswap_pwr1_input`

Get LC A2D Voltage

Node name	`$bsp_path/environment/lc<index>_a2d_volt<idx>_input`
Description	Get LC<index> A2D voltage reading
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 A2D voltage reading: `cat $bsp_path/environment/lc1_a2d_volt1_input`

Get LC A2D Voltage Scale

Node name	`$bsp_path/environment/lc<index>_a2d_volt<idx>_scale`
Description	Get LC<index> A2D voltage scale
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 A2D voltage scale: `cat $bsp_path/environment/lc1_a2d_volt1_scale`

LC LED

> **Common LED API reference:** LED_Control_API.md — line card paths under `$bsp_path/lc<N>/led/`.

Get LC Status LED

Node name	`$bsp_path/led/lc<index>_status_led`
Description	Get LC<index> status LED state
Values	lc, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get LC1 status LED: `cat $bsp_path/led/lc1_status_led`

Get LC Status LED Capabilities

Node name	`$bsp_path/led/lc<index>_status_led_capability`
Description	Get LC<index> status LED capabilities
Values	lc, Integer
Access	Read
Release version	V.7.0030.4000
Example	Get LC1 status LED capabilities: `cat $bsp_path/led/lc1_status_led_capability`

Set LC Status Green/Orange

Node name	`$bsp_path/led/lc<index>_status_led`		
Description	Set LC<index> status LED to green or orange		
Access	Write		
Release version	V.7.0030.4000		
	Name	**Data type**	**Values**
Arguments	lc	Integer	
	color	String	green, orange
Example	Set LC1 status LED to green: `echo green > $bsp_path/led/lc1_status_led`		

Set LC Status LED Green/Orange Delay Off

Node name	`$bsp_path/led/lc<index>_status_led_delay_off`		
Description	Set LC<index> status LED delay off time		
Access	Write		
Release version	V.7.0030.4000		
	Name	**Data type**	**Values**
Arguments	lc	Integer	
	delay	Integer	milliseconds
Example	Set LC1 status LED delay off to 1000ms: `echo 1000 > $bsp_path/led/lc1_status_led_delay_off`		

Set LC Status LED Green/Orange Delay On

Node name	`$bsp_path/led/lc<index>_status_led_delay_on`		
Description	Set LC<index> status LED delay on time		
Access	Write		
Release version	V.7.0030.4000		
	Name	**Data type**	**Values**
Arguments	lc	Integer	
	delay	Integer	milliseconds
Example	Set LC1 status LED delay on to 500ms: `echo 500 > $bsp_path/led/lc1_status_led_delay_on`		

LC Config

Read LC CPLD Number

Node name	`$bsp_path/config/lc<index>_cpld_num`
Description	Get LC<index> CPLD number
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 CPLD number: `cat $bsp_path/config/lc1_cpld_num`

Read LC FPGA Number

Node name	`$bsp_path/config/lc<index>_fpga_num`
Description	Get LC<index> FPGA number
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 FPGA number: `cat $bsp_path/config/lc1_fpga_num`

Read LC Gearbox Number

Node name	`$bsp_path/config/lc<index>_gearbox_num`
Description	Get LC<index> gearbox number
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 gearbox number: `cat $bsp_path/config/lc1_gearbox_num`

Read LC Gearbox Manager Number

Node name	`$bsp_path/config/lc<index>_gearbox_mgr_num`
Description	Get LC<index> gearbox manager number
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 gearbox manager number: `cat $bsp_path/config/lc1_gearbox_mgr_num`

Read LC Port Number

Node name	`$bsp_path/config/lc<index>_port_num`
Description	Get LC<index> port number
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 port number: `cat $bsp_path/config/lc1_port_num`

Read LC Module Counter

Node name	`$bsp_path/config/lc<index>_module_counter`
Description	Get LC<index> module counter
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 module counter: `cat $bsp_path/config/lc1_module_counter`

UART

Node name	`$bsp_path/system/lc<index>_uart`		
Description	Get/set LC<index> UART configuration		
Access	Read/Write		
Release version	V.7.0030.4000		
Arguments	**Name**	**Data type**	**Values**
	lc	Integer	
Example	Get LC1 UART configuration: `cat $bsp_path/system/lc1_uart`		

ASIC control

Node name	`$bsp_path/system/lc<index>_asic_control`		
Description	Get/set LC<index> ASIC control		
Access	Read/Write		
Release version	V.7.0030.4000		
Arguments	**Name**	**Data type**	**Values**
	lc	Integer	
Example	Get LC1 ASIC control: `cat $bsp_path/system/lc1_asic_control`		

LC thermal

Read LC Gearbox Temperature Input

Node name	`$bsp_path/thermal/lc<index>_gearbox<idx>_temp_input`
Description	Get LC<index> gearbox<idx> temperature input
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 gearbox1 temperature: `cat $bsp_path/thermal/lc1_gearbox1_temp_input`

Get LC QSFP/SFP Module Thermal

Node name	`$bsp_path/thermal/lc<index>_module<idx>_temp_input`
Description	Get LC<index> QSFP/SFP module<idx> temperature
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 module1 temperature: `cat $bsp_path/thermal/lc1_module1_temp_input`

Read LC Temperature Critical Module

Node name	`$bsp_path/thermal/lc<index>_module<idx>_temp_crit`
Description	Get LC<index> module<idx> critical temperature threshold
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 module1 critical temperature: `cat $bsp_path/thermal/lc1_module1_temp_crit`

Read LC Temperature Emergency Module

Node name	`$bsp_path/thermal/lc<index>_module<idx>_temp_emergency`
Description	Get LC<index> module<idx> emergency temperature threshold
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 module1 emergency temperature: `cat $bsp_path/thermal/lc1_module1_temp_emergency`

Read LC Temperature Fault Module

Node name	`$bsp_path/thermal/lc<index>_module<idx>_temp_fault`
Description	Get LC<index> module<idx> temperature fault status
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 module1 temperature fault: `cat $bsp_path/thermal/lc1_module1_temp_fault`

Read LC Temperature Input Module

Node name	`$bsp_path/thermal/lc<index>_module<idx>_temp_input`
Description	Get LC<index> module<idx> temperature input
Values	lc, Integer
Access	Read only
Release version	V.7.0030.4000
Example	Get LC1 module1 temperature input: `cat $bsp_path/thermal/lc1_module1_temp_input`

LED Control

> **Common LED API reference:** LED_Control_API.md — runtime `led_<func>` naming, sysfs tree, CPLD hardware ownership, worst-status aggregation, kernel patches, and examples. Runtime layout: examples/hw-management-led-sysfs.txt.

Get Fan Status LED

Node name	`$bsp_path/led/fan<index>_status_led`
Description	Get fan<index> status LED state
Values	fan, Integer
Access	Read
Release version	1.0
Example	Get fan1 status LED: `cat $bsp_path/led/fan1_status_led`

Get Fan LED Capabilities

Node name	`$bsp_path/led/fan<index>_status_led_capability`
Description	Get fan<index> LED capabilities
Values	fan, Integer
Access	Read
Release version	1.0
Example	Get fan1 LED capabilities: `cat $bsp_path/led/fan1_status_led_capability`

Set Fan LED Green/[Amber/Red]

Node name	`$bsp_path/led/fan<index>_status_led`		
Description	Set fan<index> LED to green or amber/red		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	fan	Integer	
	color	String	green, amber, red
Example	Set fan1 LED to green: `echo green > $bsp_path/led/fan1_status_led`		

Set Fan LED Green/[Amber/Red] Delay Off

Node name	`$bsp_path/led/fan<index>_status_led_delay_off`		
Description	Set fan<index> LED delay off time		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	fan	Integer	
	delay	Integer	milliseconds
Example	Set fan1 LED delay off to 1000ms: `echo 1000 > $bsp_path/led/fan1_status_led_delay_off`		

Set Fan LED Green/[Amber/Red] Delay On

Node name	`$bsp_path/led/fan<index>_status_led_delay_on`		
Description	Set fan<index> LED delay on time		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	fan	Integer	
	delay	Integer	milliseconds
Example	Set fan1 LED delay on to 500ms: `echo 500 > $bsp_path/led/fan1_status_led_delay_on`		

Get PSU LED

Node name	`$bsp_path/led/psu<index>_status_led`
Description	Get PSU<index> LED state
Values	psu, Integer
Access	Read
Release version	1.0
Example	Get PSU1 LED: `cat $bsp_path/led/psu1_status_led`

Get PSU LED Capabilities

Node name	`$bsp_path/led/psu<index>_status_led_capability`
Description	Get PSU<index> LED capabilities
Values	psu, Integer
Access	Read
Release version	1.0
Example	Get PSU1 LED capabilities: `cat $bsp_path/led/psu1_status_led_capability`

Set PSU LED Green/[Amber/Red]

Node name	`$bsp_path/led/psu<index>_status_led`		
Description	Set PSU<index> LED to green or amber/red		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	psu	Integer	
	color	String	green, amber, red
Example	Set PSU1 LED to green: `echo green > $bsp_path/led/psu1_status_led`		

Set PSU LED Green/[Amber/Red] Delay Off

Node name	`$bsp_path/led/psu<index>_status_led_delay_off`		
Description	Set PSU<index> LED delay off time		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	psu	Integer	
	delay	Integer	milliseconds
Example	Set PSU1 LED delay off to 1000ms: `echo 1000 > $bsp_path/led/psu1_status_led_delay_off`		

Set PSU LED Green/[Amber/Red] Delay On

Node name	`$bsp_path/led/psu<index>_status_led_delay_on`		
Description	Set PSU<index> LED delay on time		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	psu	Integer	
	delay	Integer	milliseconds
Example	Set PSU1 LED delay on to 500ms: `echo 500 > $bsp_path/led/psu1_status_led_delay_on`		

Get Status LED

Node name	`$bsp_path/led/status_led`
Description	Get system status LED state
Values	status, String
Access	Read
Release version	1.0
Example	Get system status LED: `cat $bsp_path/led/status_led`

Get Status LED Capabilities

Node name	`$bsp_path/led/status_led_capability`
Description	Get system status LED capabilities
Values	capability, String
Access	Read
Release version	1.0
Example	Get system status LED capabilities: `cat $bsp_path/led/status_led_capability`

Set Status Green/[Amber/Red]

Node name	`$bsp_path/led/status_led`		
Description	Set system status LED to green or amber/red		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	color	String	green, amber, red
Example	Set system status LED to green: `echo green > $bsp_path/led/status_led`		

Set Status LED Green/[Amber/Red] Delay Off

Node name	`$bsp_path/led/status_led_delay_off`		
Description	Set system status LED delay off time		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	delay	Integer	milliseconds
Example	Set system status LED delay off to 1000ms: `echo 1000 > $bsp_path/led/status_led_delay_off`		

Set Status LED Green/[Amber/Red] Delay On

Node name	`$bsp_path/led/status_led_delay_on`		
Description	Set system status LED delay on time		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	delay	Integer	milliseconds
Example	Set system status LED delay on to 500ms: `echo 500 > $bsp_path/led/status_led_delay_on`		

Get LED Control Owner

Stack	Host
Node name	`$bsp_path/led/led_<name>_control`
Description	Control owner of LED `<name>` (`status`, `fan`, `fan1`, `psu`, `uid`, ...). Created on LED udev add next to `led_<name>_capability` and `led_<name>_state`. Value is resolved from `$bsp_path/config/led_control_type` (Read LED Control Type Map). Missing map or unmatched name uses `led_hw_sw`. `led_sw` — Software controlled `led_hw` — Hardware controlled `led_hw_sw` — Hardware and software controlled (default)
Values	name, String: LED type from the udev LED class name; control, String: `led_sw`, `led_hw`, `led_hw_sw`
Access	Read only
Release version	3.2.7
Example	Get status LED control owner (default when no platform map): `cat $bsp_path/led/led_status_control led_hw_sw`

Power Control

Get Power Consumption

Node name	`$bsp_path/power/pwr_consum` `$bsp_path/power/pwr_sys`
Description	Get system power consumption values - `pwr_consum`: Power consumption calculation based on PSU selection - `pwr_sys`: System power consumption
Values	psu1, String: Get power consumption for PSU1; psu2, String: Get power consumption for PSU2
Access	Read only
Release version	1.0
Example	Get power consumption for PSU1: `cat $bsp_path/power/pwr_consum psu1`

Get PSU sensor Current + thresholds

Node name	`$bsp_path/power/psu<index>_curr<idx>_input`
	`$bsp_path/power/psu<index>_curr<idx>_crit`
	`$bsp_path/power/psu<index>_curr<idx>_max`

Description	Get PSU<index> current sensor readings and thresholds
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Get PSU1 current sensor: `cat $bsp_path/power/psu1_curr1_input cat $bsp_path/power/psu1_curr1_crit cat $bsp_path/power/psu1_curr1_max`

Get PSU sensor Voltage + thresholds

Node name	`$bsp_path/power/psu<index>_in<idx>_input`
	`$bsp_path/power/psu<index>_in<idx>_crit`
	`$bsp_path/power/psu<index>_in<idx>_max`

Description	Get PSU<index> voltage sensor readings and thresholds
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Get PSU1 voltage sensor: `cat $bsp_path/power/psu1_in1_input cat $bsp_path/power/psu1_in1_crit cat $bsp_path/power/psu1_in1_max`

Get PSU sensor Power + thresholds

Node name	`$bsp_path/power/psu<index>_power<idx>_input`
	`$bsp_path/power/psu<index>_power<idx>_crit`
	`$bsp_path/power/psu<index>_power<idx>_max`

Description	Get PSU<index> power sensor readings and thresholds
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Get PSU1 power sensor: `cat $bsp_path/power/psu1_power1_input cat $bsp_path/power/psu1_power1_crit cat $bsp_path/power/psu1_power1_max`

Get PSU sensor capability

Node name	`$bsp_path/power/psu<index>_sensor_capability`
Description	Get PSU<index> sensor capabilities
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Get PSU1 sensor capabilities: `cat $bsp_path/power/psu1_sensor_capability`

System / Power Control

Get ASIC Health

Node name	`$bsp_path/system/asic_health` `$bsp_path/system/asic<N>_health` (for multi-ASIC systems)
Description	Get ASIC health status. For multi-ASIC systems (N61XX_LD family: N6100_LD), additional nodes exist for each ASIC. Note: SN6600_LD (SN66XX_LD) is a single-ASIC system; only `asic_health` applies (no `asic2_health`..`asic4_health` in that configuration). Note: N6100_LD has 4 ASICs with individual health monitoring: - `asic_health` (ASIC1), `asic2_health`, `asic3_health`, `asic4_health`
Values	health, Integer: 0 (unhealthy), 1 (healthy)
Access	Read only
Release version	1.0 (V.7.0060.1000 for multi-ASIC)
Example	Get ASIC health: `# Single-ASIC systems cat $bsp_path/system/asic_health # Multi-ASIC systems (N6100_LD) cat $bsp_path/system/asic_health # ASIC1 cat $bsp_path/system/asic2_health # ASIC2 cat $bsp_path/system/asic3_health # ASIC3 cat $bsp_path/system/asic4_health # ASIC4`

MCU Reset Control

Node name	`$bsp_path/system/mcu<N>_reset`		
Description	Control MCU (Microcontroller Unit) reset for systems with multiple MCUs. Used for managing embedded controllers. Note: This attribute is for systems with MCU reset support (N61XX_LD family: N6100_LD). N6100_LD has 2 MCUs (mcu1_reset, mcu2_reset).		
Access	Read/Write		
Release version	V.7.0060.1000		
	Name	**Data type**	**Values**
Arguments	N	Integer	1-2 (N6100_LD)
	reset	Integer	0 (normal), 1 (reset)
Example	Reset MCU1: `echo 1 > $bsp_path/system/mcu1_reset cat $bsp_path/system/mcu1_reset cat $bsp_path/system/mcu2_reset`		

Get CPLD Major Version

Node name	`$bsp_path/system/cpld_major_version`
Description	Get CPLD major version
Values	version, Integer
Access	Read only
Release version	1.0
Example	Get CPLD major version: `cat $bsp_path/system/cpld_major_version`

Get CPLD Part Number

Node name	`$bsp_path/system/cpld_part_number`
Description	Get CPLD part number
Values	part_number, String
Access	Read only
Release version	1.0
Example	Get CPLD part number: `cat $bsp_path/system/cpld_part_number`

Get CPLD Minor Version

Node name	`$bsp_path/system/cpld_minor_version`
Description	Get CPLD minor version
Values	version, Integer
Access	Read only
Release version	1.0
Example	Get CPLD minor version: `cat $bsp_path/system/cpld_minor_version`

Get CPLD Full Version

Node name	`$bsp_path/system/cpld_full_version`
Description	Get CPLD full version
Values	version, String
Access	Read only
Release version	1.0
Example	Get CPLD full version: `cat $bsp_path/system/cpld_full_version`

Fan Direction

Node name	`$bsp_path/system/fan_direction`		
Description	Get/set fan direction		
Access	Read/Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	direction	String	intake, exhaust
Example	Get fan direction: `cat $bsp_path/system/fan_direction`		

Set JTAG Mode

Node name	`$bsp_path/system/jtag_mode`		
Description	Set JTAG mode		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	mode	Integer	0/1
Example	Set JTAG mode: `echo 1 > $bsp_path/system/jtag_mode`		

Set PSU On/Off

Node name	`$bsp_path/system/psu<index>_on_off`		
Description	Set PSU<index> on/off		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	psu	Integer	
	state	Integer	0/1
Example	Set PSU1 on: `echo 1 > $bsp_path/system/psu1_on_off`		

Set System Power Cycle

Node name	`$bsp_path/system/power_cycle`		
Description	Set system power cycle		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	cycle	Integer	0/1
Example	Set system power cycle: `echo 1 > $bsp_path/system/power_cycle`		

Set System Power Down

Node name	`$bsp_path/system/power_down`		
Description	Set system power down		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	power	Integer	0/1
Example	Set system power down: `echo 1 > $bsp_path/system/power_down`		

CPU Shut Down Request

Node name	`$bsp_path/system/cpu_shutdown_req`		
Description	CPU initiated shutdown request signal. This attribute is used by the CPU to request a system shutdown sequence. On liquid-cooled platforms where `hw_management_platform_config.py` defines a monitor entry, hw-mgmt polls this node (not `events/graceful_pwr_off`) and invokes `run_power_button_event` when the value changes. Platforms with this entry in the current tree include HI162, HI166–HI170, HI176, HI177, HI180, HI185 (N6300_LD), and others listed in `PLATFORM_CONFIG`.		
Access	Read/Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	request	Integer	0/1
Example	Set CPU shutdown request: `echo 1 > $bsp_path/system/cpu_shutdown_req`		

Graceful Power Off

Node name	`$bsp_path/system/graceful_power_off`		
Description	Initiate graceful system power off sequence. This attribute triggers a controlled shutdown procedure that allows the system to properly close processes and save state before powering off.		
Access	Read/Write		
Release version	3.0		
Arguments	**Name**	**Data type**	**Values**
	poweroff	Integer	0/1
Example	Initiate graceful power off: `echo 1 > $bsp_path/system/graceful_power_off`		
Note	See also Graceful Power Off Event for the related event `graceful_pwr_off` that monitors graceful shutdown status.		

CPU Power off Ready

Node name	`$bsp_path/system/cpu_power_off_ready`		
Description	CPU signal indicating readiness for power off. This attribute is set by the CPU to signal that it has completed shutdown procedures and is ready for the system to power down.		
Access	Read/Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	ready	Integer	0/1
Example	Set CPU power off ready status: `echo 1 > $bsp_path/system/cpu_power_off_ready`		

Shutdown Unlock

Node name	`$bsp_path/system/shutdown_unlock`		
Description	Unlock mechanism for shutdown operations. This attribute must be set to enable shutdown and power cycle operations, providing a safety mechanism to prevent accidental system shutdowns.		
Access	Read/Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	unlock	Integer	0/1
Example	Unlock shutdown operations: `echo 1 > $bsp_path/system/shutdown_unlock`		
Example	Lock shutdown operations: `echo 0 > $bsp_path/system/shutdown_unlock`		

Boot Completed

Node name	`$bsp_path/system/boot_completed`		
Description	System boot completion status. This attribute indicates that the system has completed its boot sequence and is fully operational.		
Access	Read/Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	status	Integer	0 (not completed), 1 (completed)
Example	Read boot completion status: `cat $bsp_path/system/boot_completed`		
Example	Set boot completed flag: `echo 1 > $bsp_path/system/boot_completed`		
Note	On AMD SNW CPU platforms, this may be implemented as a GPIO-based attribute.		

Set Line Card Power

Node name	`$bsp_path/system/lc<index>_power`		
Description	Set line card<index> power		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	lc	Integer	
	power	Integer	0/1
Example	Set LC1 power: `echo 1 > $bsp_path/system/lc1_power`		

Set Line Card Enable

Node name	`$bsp_path/system/lc<index>_enable`		
Description	Set line card<index> enable		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	lc	Integer	
	enable	Integer	0/1
Example	Set LC1 enable: `echo 1 > $bsp_path/system/lc1_enable`		

Read Line Card Active

Node name	`$bsp_path/system/lc<index>_active`
Description	Read line card<index> active status
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 active status: `cat $bsp_path/system/lc1_active`

Read Line Card Powered

Node name	`$bsp_path/system/lc<index>_powered`
Description	Read line card<index> powered status
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 powered status: `cat $bsp_path/system/lc1_powered`

Read Line Card Present

Node name	`$bsp_path/system/lc<index>_present`
Description	Read line card<index> present status
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 present status: `cat $bsp_path/system/lc1_present`

Read Line Card Ready

Node name	`$bsp_path/system/lc<index>_ready`
Description	Read line card<index> ready status
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 ready status: `cat $bsp_path/system/lc1_ready`

Read Line Card Synced

Node name	`$bsp_path/system/lc<index>_synced`
Description	Read line card<index> synced status
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 synced status: `cat $bsp_path/system/lc1_synced`

Read Line Card Verified

Node name	`$bsp_path/system/lc<index>_verified`
Description	Read line card<index> verified status
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 verified status: `cat $bsp_path/system/lc1_verified`

Read Line Card Reset Mask

Node name	`$bsp_path/system/lc<index>_reset_mask`
Description	Read line card<index> reset mask
Values	lc, Integer
Access	Read only
Release version	1.0
Example	Read LC1 reset mask: `cat $bsp_path/system/lc1_reset_mask`

Set Line Card Shutdown

Node name	`$bsp_path/system/lc<index>_shutdown`		
Description	Set line card<index> shutdown		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	lc	Integer	
	shutdown	Integer	0/1
Example	Set LC1 shutdown: `echo 1 > $bsp_path/system/lc1_shutdown`		

Set VPD Write Protect

Node name	`$bsp_path/system/vpd_write_protect`		
Description	Set VPD write protect		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	protect	Integer	0/1
Example	Set VPD write protect: `echo 1 > $bsp_path/system/vpd_write_protect`		

Set ASIC Up during PCIe root complex reset

Node name	`$bsp_path/system/asic_up_during_pcie_reset`		
Description	Set ASIC up during PCIe root complex reset		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	asic	Integer	0/1
Example	Set ASIC up during PCIe reset: `echo 1 > $bsp_path/system/asic_up_during_pcie_reset`		

Get Voltreg Update status

Node name	`$bsp_path/system/voltreg_update_status`
Description	Get voltage regulator update status
Values	status, String
Access	Read only
Release version	1.0
Example	Get voltage regulator update status: `cat $bsp_path/system/voltreg_update_status`

Get Config1, Config2

Node name	`$bsp_path/system/config1` `$bsp_path/system/config2`
Description	Get system configuration 1 and 2
Values	config, Integer
Access	Read only
Release version	1.0
Example	Get system configurations: `cat $bsp_path/system/config1 cat $bsp_path/system/config2`

Get Ufm Version

Node name	`$bsp_path/system/ufm_version`
Description	Get UFM version
Values	version, String
Access	Read only
Release version	1.0
Example	Get UFM version: `cat $bsp_path/system/ufm_version`

DPU Power Off

Node name	`$bsp_path/system/dpu<index>_power_off`		
Description	Set DPU<index> power off		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	dpu	Integer	
	power	Integer	0/1
Example	Set DPU1 power off: `echo 1 > $bsp_path/system/dpu1_power_off`		

DPU Force Power Off

Node name	`$bsp_path/system/dpu<index>_force_power_off`		
Description	Force DPU<index> power off		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	dpu	Integer	
	force	Integer	0/1
Example	Force DPU1 power off: `echo 1 > $bsp_path/system/dpu1_force_power_off`		

DPU Reset

Node name	`$bsp_path/system/dpu<index>_reset`		
Description	Set DPU<index> reset		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	dpu	Integer	
	reset	Integer	0/1
Example	Set DPU1 reset: `echo 1 > $bsp_path/system/dpu1_reset`		

DPU Reset Enable

Node name	`$bsp_path/system/dpu<index>_reset_enable`		
Description	Enable DPU<index> reset		
Access	Write		
Release version	1.0		
	Name	**Data type**	**Values**
Arguments	dpu	Integer	
	enable	Integer	0/1
Example	Enable DPU1 reset: `echo 1 > $bsp_path/system/dpu1_reset_enable`		

Get Reset Cause

Node name	`$bsp_path/system/reset_<cause>`
Description	Get last system reset cause. Reset causes vary between platform families. Each supported cause is exposed as a separate read-only attribute under `$bsp_path/system/`. Reading `$bsp_path/system/reset_cause` returns the primary cause string. For most causes only one attribute reads `1`; the rest read `0`.
Values	cause, String: Primary reset cause (via `reset_cause`); `reset_<cause>`, Integer: `1` = this cause triggered the reset, `0` = not related
Access	Read only
Release version	1.0
Example	Check if long button press caused reset: `cat $bsp_path/system/reset_long_pb`
Example	Get reset cause string: `cat $bsp_path/system/reset_cause`

N51XX_LD platform family

N51XX_LD platforms do **not** support `reset_ac_pwr_fail`, `reset_aux_pwr_or_ref`, `reset_comex_wd`, `reset_from_asic`, `reset_from_comex`, or `reset_reload_bios`.

Applies to N5110_LD, N5112_LD, N5100_LD, N5101_LD, N5200_LD, N5201_LD, N5300_LD, N5400_LD, N5120_LD, N5121_LD, N5320_LD, N5500_LD (GB200/GB300), and N5240_LD (Kyber). `config/reset_attr_num` = 22.

Reset cause attribute	Description
`reset_long_pb`	Reset button was pushed for more than 15 seconds
`reset_short_pb`	Reset button was pushed for less than 15 seconds
`reset_aux_pwr_or_fu`	Reset due to CPLD power down or CPLD code refresh
`reset_swb_dc_dc_pwr_fail`	Switch board DC/DC power failure
`reset_sw_reset`	Power cycle command initiated by software
`reset_pwr_button_or_leak_con`	Reset triggered by power button or leak connector event
`reset_swb_wd`	Reset or power off initiated by switch-board watchdog
`reset_asic_thermal`	ASIC power drop due to failure or thermal shutdown
`reset_cpu_thermal`	CPU thermal shutdown
`reset_aux_pwr_or_reload`	Auxiliary power failure or CPLD field upgrade
`reset_comex_pwr_fail`	COMe module power failure
`reset_platform`	Reboot command initiated by software
`reset_soc`	Power off initiated by SoC (for example Linux `poweroff`)
`reset_from_erot`	Reset initiated by eRoT
`reset_erot`	eRoT reset

Reset cause attribute	Description
`reset_system`	System reset cycle, power on, power cycle, or ASIC reset
`reset_sw_pwr_off`	Power off initiated by software through CPLD
`reset_comex_thermal`	COMe thermal shutdown
`reset_comex_power`	COMe power event
`reset_pwr_converter_fail`	Power converter failure
`reset_main_5v`	Management board 5 V rail failure
`reset_mgmt_pwr`	Management board power failure

Validation source: `recipes-kernel/linux/linux-6.12/9007-platform-mellanox-Downstream-Introduce-support-of-Nv.patch`, `usr/usr/bin/hw-management.sh` (`n51xx_reset_attr_num`).

SN5610 / SN5640

SN5610 (HI172) and SN5640 (HI171) expose `reset_reload_bios` as a read-only status attribute, but the CPLD does not implement the trigger that would set it, so this reset cause cannot be produced on these systems.

Validation source: `mlxplat_mlxcpld_default_ng_regs_io_data` in the mlx-platform driver (`.mode = 0444`).

SN6810_LD / SN6800_LD

SN6810_LD (HI183) and SN6800_LD (HI187) do not support triggering `reset_system`.

Cartridge

Node name	`$bsp_path/system/cartridge<index>`
Description	Get cable cartridge status at the specified index. Cable cartridges are used in liquid-cooled multi-ASIC systems (N61XX_LD family: N6100_LD, N6300_LD) for connecting external cables. Note: This attribute is primarily for liquid-cooled systems with cable cartridges (N61XX_LD family: N6100_LD, N6300_LD). N6100_LD and N6300_LD (SKU HI185) have 4 cable cartridges.
Values	index, Integer: 1-4 (N6100_LD, N6300_LD); Status, Integer: 0/1
Access	Read only
Release version	V.7.0060.1000
Example	Get cartridge 1 status: `cat $bsp_path/system/cartridge1`

ASIC PG Failure

Node name	`$bsp_path/system/asic_pg_fail`
Description	Get ASIC power good failure status. Indicates power sequencing failure on the ASIC. Note: This attribute is for multi-ASIC systems (SN58XX_LD family, N61XX_LD family). N6100_LD has 4 ASICs with individual failure tracking. Exception: This API does not exist on SN6600_LD, SN6800_LD, and SN6810_LD systems.
Values	Status, Integer: 0/1
Access	Read only
Release version	V.7.0060.1000
Example	Get ASIC power good failure status: `cat $bsp_path/system/asic_pg_fail`

Leakage Sensors

Leakage Sensor

Node name	`$bsp_path/system/leakage<index>`
Description	Get leakage sensor status at the specified index. Detects liquid leaks in liquid-cooled systems. Note: This attribute is for liquid-cooled systems (SN58XX_LD family, N61XX_LD family, SN66XX_LD family). N6100_LD has 2 leakage sensors, SN5810_LD has 2, SN5800_LD has 5. N6300_LD (SKU HI185): `config/leakage_counter` is **2** (`n61xxld_specific()` / HI185 branch in `hw-management.sh`). SN6600_LD: `config/leakage_counter` is **2**; the validated sysfs tree may still list `system/leakage1` through `system/leakage5` when the hardware exposes those mlxreg-io channels (see the `leakage_counter` section note above).
Values	index, Integer: 1-X (system dependent); Status, Integer: 0=no leak, 1=leak detected
Access	Read only
Release version	V.7.0040.3930
Example	Get leakage sensor 1 status: `cat $bsp_path/system/leakage1`

Thermal

Reference — Host: `usr/usr/bin/hw-management-thermal-events.sh`; platform sensors in `usr/etc/hw-management-sensors/`.

Reference — BMC (HI189): `bmc/usr/etc/HI189/hw-management-bmc-events.sh`; `bmc/examples/hw-management-bmc-thermal-sysfs.txt`.

Ambient sensors

Stack	Host (`hw-management-thermal-events.sh`)
Node name	`$bsp_path/thermal/<ambient_name>` (platform-specific; examples include `mng_amb`, `comex_amb`, `port_amb`, `fan_amb`, `swb_amb`, `cpu_amb`)
Description	Ambient temperature sensors are created by `hw-management-thermal-events.sh` when the corresponding hwmon device appears. The stable symlink name under `$bsp_path/thermal/` depends on the platform label and sensor type. Some platforms also expose management ambient via `$bsp_path/thermal/mng_amb` (see **MNG Temperature**). Legacy documentation referenced `$bsp_path/system/amb_sens` for mlxreg ambient enablement on certain systems.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	Read management ambient temperature (when present): `cat $bsp_path/thermal/mng_amb`

Read Switch ASIC Temperature

Node name	`$bsp_path/thermal/asic<index>_temp_input`
Description	Read switch ASIC<index> temperature
Values	asic, Integer
Access	Read only
Release version	1.0
Example	Read ASIC1 temperature: `cat $bsp_path/thermal/asic1_temp_input`

Read Switch ASIC Temperature Normal

Node name	`$bsp_path/thermal/asic<index>_temp_normal`
Description	Read switch ASIC<index> normal temperature threshold
Values	asic, Integer
Access	Read only
Release version	1.0
Example	Read ASIC1 normal temperature: `cat $bsp_path/thermal/asic1_temp_normal`

Read Switch ASIC Temperature Critical

Node name	`$bsp_path/thermal/asic<index>_temp_crit`
Description	Read switch ASIC<index> critical temperature threshold
Values	asic, Integer
Access	Read only
Release version	1.0
Example	Read ASIC1 critical temperature: `cat $bsp_path/thermal/asic1_temp_crit`

Read Switch ASIC Temperature Emergency

Node name	`$bsp_path/thermal/asic<index>_temp_emergency`
Description	Read switch ASIC<index> emergency temperature threshold
Values	asic, Integer
Access	Read only
Release version	1.0
Example	Read ASIC1 emergency temperature: `cat $bsp_path/thermal/asic1_temp_emergency`

Read Switch ASIC Temperature Trip Critical

Node name	`$bsp_path/thermal/asic<index>_temp_trip_crit`
Description	Read switch ASIC<index> trip critical temperature
Values	asic, Integer
Access	Read only
Release version	1.0
Example	Read ASIC1 trip critical temperature: `cat $bsp_path/thermal/asic1_temp_trip_crit`

Read Switch Comex Temperature

Node name	`$bsp_path/thermal/comex_temp_input`
Description	Read switch Comex temperature
Values	temperature, Integer
Access	Read only
Release version	1.0
Example	Read Comex temperature: `cat $bsp_path/thermal/comex_temp_input`

MNG Temperature

Stack	Host (`hw-management-thermal-events.sh`)
Node name	`$bsp_path/thermal/mng_amb`
Description	Read management (MNG) ambient temperature when the platform exposes this sensor through the hw-mgmt thermal events handler.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	Read MNG ambient temperature: `cat $bsp_path/thermal/mng_amb`

Read host CPU temperature (BMC stack)

Stack	BMC (`hw-management-bmc-events.sh`, event `cpu_temp`)
Node name	`$bsp_path/thermal/cpu_temp_input`
Description	Instantaneous **host CPU** temperature as seen on the **BMC** image. The BMC early-I2C map binds bus 15 / `0x4c` to the **sbtsi** driver (`hw-management-bmc-early-i2c-devices.json`). This is not created by the host thermal events script on the CPU image.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/cpu_temp_input`

Read host CPU temperature max / min (BMC stack)

Stack	BMC (`hw-management-bmc-events.sh`, event `cpu_temp`)
Node name	`$bsp_path/thermal/cpu_temp` (maps to hwmon `temp1_max`); `$bsp_path/thermal/cpu_min` (maps to hwmon `temp1_min`)
Description	High/low limit attributes from the same **sbtsi** sensor as `cpu_temp_input`. **sbtsi** exposes `temp1_min`; both symlinks are created on HI189.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/cpu_temp cat $bsp_path/thermal/cpu_min`

Read BMC Temperature

Stack	BMC (`hw-management-bmc-events.sh`, event `bmc_temp`)
Node name	`$bsp_path/thermal/bmc_temp_input`
Description	Read BMC ambient temperature (instantaneous reading from the BMC hwmon sensor). Symlink is created under `/var/run/hw-management/thermal/` by `hw-management-bmc-events.sh` when the BMC temperature hwmon device appears (for example HI189 / SN6600: I2C `4-0048`, `lm75` driver, `temp1_input`).
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Note	Supported on systems with the hw-mgmt BMC thermal stack only. Values are typically millidegrees Celsius (Linux hwmon convention).
Example	Read BMC ambient temperature: `cat $bsp_path/thermal/bmc_temp_input`

Read BMC Max Temperature

Stack	BMC (`hw-management-bmc-events.sh`, event `bmc_temp` — **BMC board** lm75 sensor, not host CPU)
Node name	`$bsp_path/thermal/bmc_temp`
Description	Read BMC ambient high-temperature limit (maps to hwmon `temp1_max` on the BMC sensor). Created by the same BMC thermal udev handler as `bmc_temp_input`. On HI189 the BMC sensor uses the `lm75` driver, which exposes `temp1_input` and `temp1_max` only (no `temp1_min`).
Values	temperature, Integer: driver-specific limit semantics
Access	Read only
Release version	1.0
Note	Supported on systems with the hw-mgmt BMC thermal stack only. Do not expect `thermal/bmc_min` on `lm75`-backed BMC sensors; the driver does not register a minimum temperature attribute, so `hw-management-bmc-events.sh` does not create that symlink.
Example	Read BMC max temperature threshold: `cat $bsp_path/thermal/bmc_temp`

Read BMC Critical Temperature

Stack	BMC — **not created** on HI189 (`lm75` has no `temp1_crit`)
Node name	`$bsp_path/thermal/bmc_crit` (legacy); not created on HI189 BMC stack
Description	On systems with the hw-mgmt BMC thermal stack (HI189), the BMC ambient sensor uses the `lm75` driver, which does not expose `temp1_crit` in sysfs. `hw-management-bmc-events.sh` does not create a `bmc_crit` symlink. Rev. 2.8 documented `$bsp_path/thermal/bmc_crit`; use `bmc_temp` (max/limit) and `bmc_temp_input` on current BMC platforms.
Access	Read only (when node exists on other platforms)
Release version	1.0
Note	Virtual test trees (for example HI193) may still ship fixture files named `bmc_crit` under `usr/etc/hw-management-virtual/`; that naming does not apply to live HI189 runtime symlinks.
Example	Not applicable on HI189 — verify with `ls $bsp_path/thermal/` on the device.

Read BMC Minimal Temperature

Stack	BMC — **not created** on HI189 (`lm75` has no `temp1_min`; `check_n_link` skips)
Node name	`$bsp_path/thermal/bmc_min` (legacy); not created on HI189 BMC stack
Description	The BMC events handler attempts `temp1_min`, then `bmc_min`, but `check_n_link` silently skips missing sources. The HI189 BMC sensor (`lm75` at I2C `4-0048`) has no `temp1_min` attribute, so **'bmc_min' never appears** on the BMC stack using this stack. See `bmc/examples/hw-management-bmc-thermal-sysfs.txt`.
Access	Read only (when node exists on other platforms)
Release version	1.0
Example	Not applicable on HI189: `# Expected absent on lm75-backed BMC: ls $bsp_path/thermal/bmc_min`

Read Cooling State

Node name	`$bsp_path/thermal/cooling_state`
Description	Read system cooling state
Values	state, String
Access	Read only
Release version	1.0
Example	Read cooling state: `cat $bsp_path/thermal/cooling_state`

Cooling Name

Node name	`$bsp_path/config/cooling_name`
Description	Thermal control cooling device name written by `hw-management-thermal-events.sh` during cooling device registration. Used by the thermal control daemon configuration.
Values	name, String: platform-specific
Access	Read only
Release version	1.0
Example	Read cooling name: `cat $bsp_path/config/cooling_name`

Read CPU Core Temperature

Node name	`$bsp_path/thermal/cpu_core<index>_temp_input`
Description	Read CPU core<index> temperature
Values	core, Integer
Access	Read only
Release version	1.0
Example	Read CPU core1 temperature: `cat $bsp_path/thermal/cpu_core1_temp_input`

CPU Core Critical Temperature

Node name	`$bsp_path/thermal/cpu_core<index>_temp_crit`
Description	Read CPU core<index> critical temperature threshold
Values	core, Integer
Access	Read only
Release version	1.0
Example	Read CPU core1 critical temperature: `cat $bsp_path/thermal/cpu_core1_temp_crit`

CPU Core Critical Temperature Alarm

Node name	`$bsp_path/thermal/cpu_core<index>_temp_crit_alarm`
Description	Read CPU core<index> critical temperature alarm
Values	core, Integer
Access	Read only
Release version	1.0
Example	Read CPU core1 critical temperature alarm: `cat $bsp_path/thermal/cpu_core1_temp_crit_alarm`

CPU Core Temperature Max

Node name	`$bsp_path/thermal/cpu_core<index>_temp_max`
Description	Read CPU core<index> maximum temperature
Values	core, Integer
Access	Read only
Release version	1.0
Example	Read CPU core1 maximum temperature: `cat $bsp_path/thermal/cpu_core1_temp_max`

Read CPU Pack Temperature

Node name	`$bsp_path/thermal/cpu_pack_temp_input`
Description	Read CPU pack temperature
Values	temperature, Integer
Access	Read only
Release version	1.0
Example	Read CPU pack temperature: `cat $bsp_path/thermal/cpu_pack_temp_input`

CPU Pack Critical Temperature

Node name	`$bsp_path/thermal/cpu_pack_temp_crit`
Description	Read CPU pack critical temperature threshold
Values	temperature, Integer
Access	Read only
Release version	1.0
Example	Read CPU pack critical temperature: `cat $bsp_path/thermal/cpu_pack_temp_crit`

CPU Pack Critical Temperature Alarm

Node name	`$bsp_path/thermal/cpu_pack_temp_crit_alarm`
Description	Read CPU pack critical temperature alarm
Values	alarm, Integer
Access	Read only
Release version	1.0
Example	Read CPU pack critical temperature alarm: `cat $bsp_path/thermal/cpu_pack_temp_crit_alarm`

CPU Pack Temperature Max

Node name	`$bsp_path/thermal/cpu_pack_temp_max`
Description	Read CPU pack maximum temperature
Values	temperature, Integer
Access	Read only
Release version	1.0
Example	Read CPU pack maximum temperature: `cat $bsp_path/thermal/cpu_pack_temp_max`

Read Fan Max Speed

Node name	`$bsp_path/thermal/fan<index>_max_speed`
Description	Read fan<index> maximum speed
Values	fan, Integer
Access	Read only
Release version	1.0
Example	Read fan1 maximum speed: `cat $bsp_path/thermal/fan1_max_speed`

Read Fan Min Speed

Node name	`$bsp_path/thermal/fan<index>_min_speed`
Description	Read fan<index> minimum speed
Values	fan, Integer
Access	Read only
Release version	1.0
Example	Read fan1 minimum speed: `cat $bsp_path/thermal/fan1_min_speed`

Set Fan Speed

Node name	`$bsp_path/thermal/fan<index>_speed_set`		
Description	Set fan PWM/speed for fan `<index>`. Symlink is created by `hw-management-thermal-events.sh` from the cooling device `pwm1` attribute when supported.		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	speed	Integer	platform-specific PWM scale
Example	Set fan1 speed: `echo 50 > $bsp_path/thermal/fan1_speed_set`		

Fan Speed Tolerance (per fan)

Node name	`$bsp_path/thermal/fan<index>_speed_tolerance`
Description	Fan speed tolerance for fan `<index>` when exposed under the thermal hierarchy. For system-wide tolerance configuration see also `$bsp_path/config/fan_speed_tolerance` (Fan Speed Tolerance).
Values	tolerance, Integer
Access	Read only
Release version	1.0
Example	Read fan1 speed tolerance (when present): `cat $bsp_path/thermal/fan1_speed_tolerance`

Read Fan Direction

Node name	`$bsp_path/thermal/fan<index>_dir`
Description	Read fan drawer `<index>` airflow direction.
Access	Read only
Release version	1.0
Values	0 — Reverse (C2P) 1 — Forward (P2C) 2 — Direction cannot be determined (fan missing, or fan present but direction debounce failed) **Notes:** ● If the fan is not present (`thermal/fan<index>_status` is 0), `fan<index>_dir` is irrelevant and must be ignored by consumers. ● On systems that report direction via CPLD (`system/fan_dir`), a remove event keeps `fan<index>_dir` present and sets it to 2. ● On systems with fan-module EEPROM direction detection (`config/fan_dir_eeprom`), direction is taken from EEPROM VPD data. When the fan (and its EEPROM) is removed, `fan<index>_dir` may also be removed.
Example	Read fan1 direction: `cat $bsp_path/thermal/fan1_dir`

Read Fan Status

Node name	`$bsp_path/thermal/fan<index>_status`
Description	Read fan<index> status
Values	fan, Integer
Access	Read only
Release version	1.0
Example	Read fan1 status: `cat $bsp_path/thermal/fan1_status`

Read Fan Fault

Node name	`$bsp_path/thermal/fan<index>_fault`
Description	Read fan<index> fault status
Values	fan, Integer
Access	Read only
Release version	1.0
Example	Read fan1 fault status: `cat $bsp_path/thermal/fan1_fault`

Comex Voltmon Temperature

Node name	`$bsp_path/thermal/comex_voltmon<index>_temp<index>_input`
Description	Read Comex PMBus voltmon temperature input.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	Read comex voltmon1 temperature: `cat $bsp_path/thermal/comex_voltmon1_temp1_input`

Comex Voltmon Critical Temperature

Node name	`$bsp_path/thermal/comex_voltmon<index>_temp<index>_crit`
Description	Read Comex voltmon critical temperature threshold.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/comex_voltmon1_temp1_crit`

Comex Voltmon Max Temperature

Node name	`$bsp_path/thermal/comex_voltmon<index>_temp<index>_max`
Description	Read Comex voltmon maximum temperature threshold.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/comex_voltmon1_temp1_max`

Read Port Ambient

Node name	`$bsp_path/thermal/port<index>_ambient_temp`
Description	Read port<index> ambient temperature
Values	port, Integer
Access	Read only
Release version	1.0
Example	Read port1 ambient temperature: `cat $bsp_path/thermal/port1_ambient_temp`

Read PSU Temperature

Node name	`$bsp_path/thermal/psu<index>_temp_input`
Description	Read PSU<index> temperature
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 temperature: `cat $bsp_path/thermal/psu1_temp_input`

Read PSU Alarm

Node name	`$bsp_path/thermal/psu<index>_alarm`
Description	Read PSU<index> alarm status
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 alarm: `cat $bsp_path/thermal/psu1_alarm`

Read PSU Max

Node name	`$bsp_path/thermal/psu<index>_temp_max`
Description	Read PSU<index> maximum temperature
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 maximum temperature: `cat $bsp_path/thermal/psu1_temp_max`

Read PSU Fan Speed

Node name	`$bsp_path/thermal/psu<index>_fan_speed`
Description	Read PSU<index> fan speed
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 fan speed: `cat $bsp_path/thermal/psu1_fan_speed`

Read PSU min/max Fan Speed

Node name	`$bsp_path/thermal/psu<index>_fan_min_speed` `$bsp_path/thermal/psu<index>_fan_max_speed`
Description	Read PSU<index> minimum and maximum fan speed
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 fan speeds: `cat $bsp_path/thermal/psu1_fan_min_speed` `cat $bsp_path/thermal/psu1_fan_max_speed`

Read PSU Power Status

Node name	`$bsp_path/thermal/psu<index>_power_status`
Description	Read PSU<index> power status
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 power status: `cat $bsp_path/thermal/psu1_power_status`

Read PSU Status

Node name	`$bsp_path/thermal/psu<index>_status`
Description	Read PSU<index> status
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read PSU1 status: `cat $bsp_path/thermal/psu1_status`

Read PDB Hotswap Controller Temperature

Node name	`$bsp_path/thermal/pdb_hotswap<index>_temp<index>_input`
Description	Read PDB (Power Distribution Board) hot-swap controller temperature Note: This attribute is for liquid-cooled systems only (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD; SN66XX_LD family: SN6600_LD)
Values	Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0050.3000
Example	Read PDB hotswap 1 temperature: `cat $bsp_path/thermal/pdb_hotswap1_temp1_input`

Read PDB Hotswap Controller Temperature Thresholds

Node name	`$bsp_path/thermal/pdb_hotswap<index>_temp<index>_crit` `$bsp_path/thermal/pdb_hotswap<index>_temp<index>_max`
Description	Read PDB hot-swap controller temperature critical and maximum thresholds Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0050.3000
Example	Read PDB hotswap 1 temperature thresholds: `cat $bsp_path/thermal/pdb_hotswap1_temp1_crit cat $bsp_path/thermal/pdb_hotswap1_temp1_max`

Read PDB Power Converter Temperature

Node name	`$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_input`
Description	Read PDB power converter temperature Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0050.3000
Example	Read PDB power converter 1 temperature: `cat $bsp_path/thermal/pdb_pwr_conv1_temp1_input`

Read PDB Power Converter Temperature Thresholds

Node name	`$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_crit`
	`$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_lcrit`
	`$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_max`

Description	Read PDB power converter temperature thresholds (critical, lower critical, maximum) Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0050.3000
Example	Read PDB power converter 1 temperature thresholds: `cat $bsp_path/thermal/pdb_pwr_conv1_temp1_crit cat $bsp_path/thermal/pdb_pwr_conv1_temp1_lcrit cat $bsp_path/thermal/pdb_pwr_conv1_temp1_max`

Read PDB MOSFET Ambient Temperature

Node name	`$bsp_path/thermal/pdb_mosfet_amb<index>`
Description	Read PDB MOSFET ambient temperature sensor Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)
Values	Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0050.3000
Example	Read PDB MOSFET ambient temperature: `cat $bsp_path/thermal/pdb_mosfet_amb1`

Read System PWM1

Node name	`$bsp_path/thermal/system_pwm1`
Description	Read system PWM1 value
Values	pwm, Integer
Access	Read only
Release version	1.0
Example	Read system PWM1: `cat $bsp_path/thermal/system_pwm1`

Read Temperature Critical Module

Node name	`$bsp_path/thermal/module<index>_temp_crit`
Description	Read module<index> critical temperature threshold
Values	module, Integer
Access	Read only
Release version	1.0
Example	Read module1 critical temperature: `cat $bsp_path/thermal/module1_temp_crit`

Read Temperature Emergency Module

Node name	`$bsp_path/thermal/module<index>_temp_emergency`
Description	Read module<index> emergency temperature threshold
Values	module, Integer
Access	Read only
Release version	1.0
Example	Read module1 emergency temperature: `cat $bsp_path/thermal/module1_temp_emergency`

Read Temperature Trip Critical Module

Node name	`$bsp_path/thermal/module<index>_temp_trip_crit`
Description	Read module<index> trip critical temperature
Values	module, Integer
Access	Read only
Release version	1.0
Example	Read module1 trip critical temperature: `cat $bsp_path/thermal/module1_temp_trip_crit`

Read Temperature Fault Module

Node name	`$bsp_path/thermal/module<index>_temp_fault`
Description	Read module<index> temperature fault status
Values	module, Integer
Access	Read only
Release version	1.0
Example	Read module1 temperature fault: `cat $bsp_path/thermal/module1_temp_fault`

Read Temperature Input Module

Node name	`$bsp_path/thermal/module<index>_temp_input`
Description	Read module<index> temperature input
Values	module, Integer
Access	Read only
Release version	1.0
Example	Read module1 temperature input: `cat $bsp_path/thermal/module1_temp_input`

Read Temperature Critical Gearbox

Node name	`$bsp_path/thermal/gearbox<index>_temp_crit`
Description	Read gearbox<index> critical temperature threshold
Values	gearbox, Integer
Access	Read only
Release version	1.0
Example	Read gearbox1 critical temperature: `cat $bsp_path/thermal/gearbox1_temp_crit`

Read Temperature Emergency Gearbox

Node name	`$bsp_path/thermal/gearbox<index>_temp_emergency`
Description	Read gearbox<index> emergency temperature threshold
Values	gearbox, Integer
Access	Read only
Release version	1.0
Example	Read gearbox1 emergency temperature: `cat $bsp_path/thermal/gearbox1_temp_emergency`

Read Temperature Trip Critical Gearbox

Node name	`$bsp_path/thermal/gearbox<index>_temp_trip_crit`
Description	Read gearbox<index> trip critical temperature
Values	gearbox, Integer
Access	Read only
Release version	1.0
Example	Read gearbox1 trip critical temperature: `cat $bsp_path/thermal/gearbox1_temp_trip_crit`

Read Temperature Input Gearbox

Node name	`$bsp_path/thermal/gearbox<index>_temp_input`
Description	Read gearbox<index> temperature input
Values	gearbox, Integer
Access	Read only
Release version	1.0
Example	Read gearbox1 temperature input: `cat $bsp_path/thermal/gearbox1_temp_input`

Read Switch CPU Temperature

Node name	`$bsp_path/thermal/switch_cpu_temp`
Description	Read switch CPU temperature
Values	temperature, Integer
Access	Read only
Release version	1.0
Example	Read switch CPU temperature: `cat $bsp_path/thermal/switch_cpu_temp`

Read Switch Fan Temperature

Node name	`$bsp_path/thermal/switch_fan_temp`
Description	Read switch fan temperature
Values	temperature, Integer
Access	Read only
Release version	1.0
Example	Read switch fan temperature: `cat $bsp_path/thermal/switch_fan_temp`

Read Switch Port Temperature

Node name	`$bsp_path/thermal/switch_port<index>_temp`
Description	Read switch port<index> temperature
Values	port, Integer
Access	Read only
Release version	1.0
Example	Read switch port1 temperature: `cat $bsp_path/thermal/switch_port1_temp`

Read Switch Power Supply Temperature

Node name	`$bsp_path/thermal/switch_psu<index>_temp`
Description	Read switch power supply<index> temperature
Values	psu, Integer
Access	Read only
Release version	1.0
Example	Read switch PSU1 temperature: `cat $bsp_path/thermal/switch_psu1_temp`

SODIMM Temperature Input

Node name	`$bsp_path/thermal/sodimm<index>_temp_input`
Description	Read SODIMM (Small Outline Dual In-line Memory Module) temperature. SODIMMs are DDR memory modules with integrated temperature sensors. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family). SN5810_LD and N6100_LD use 2 SODIMM sensors on I2C bus 2 (addresses 0x1a, 0x1b). SN6600_LD uses 2 JC42 sensors on I2C bus 10 (addresses 0x52, 0x53) per validated tree.
Values	index, Integer: 1-2 (N6100_LD, SN6600_LD); Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0060.1000
Example	Read SODIMM temperatures: `cat $bsp_path/thermal/sodimm1_temp_input # SODIMM 1 cat $bsp_path/thermal/sodimm2_temp_input # SODIMM 2`

SODIMM Critical Temperature

Node name	`$bsp_path/thermal/sodimm<index>_temp_crit`
Description	Read SODIMM critical temperature threshold. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).
Values	index, Integer: 1-2 (N6100_LD, SN6600_LD); Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0060.1000
Example	Read SODIMM critical temperatures: `cat $bsp_path/thermal/sodimm1_temp_crit cat $bsp_path/thermal/sodimm2_temp_crit`

SODIMM Max Temperature

Node name	`$bsp_path/thermal/sodimm<index>_temp_max`
Description	Read SODIMM maximum temperature threshold. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).
Values	index, Integer: 1-2 (N6100_LD, SN6600_LD); Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0060.1000
Example	Read SODIMM max temperatures: `cat $bsp_path/thermal/sodimm1_temp_max cat $bsp_path/thermal/sodimm2_temp_max`

SODIMM Min Temperature

Node name	`$bsp_path/thermal/sodimm<index>_temp_min`
Description	Read SODIMM minimum temperature threshold. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).
Values	index, Integer: 1-2 (N6100_LD, SN6600_LD); Temperature, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0060.1000
Example	Read SODIMM min temperatures: `cat $bsp_path/thermal/sodimm1_temp_min cat $bsp_path/thermal/sodimm2_temp_min`

SODIMM Temperature Alarms

Node name	`$bsp_path/thermal/sodimm<index>_temp_crit_alarm`
	`$bsp_path/thermal/sodimm<index>_temp_max_alarm`
	`$bsp_path/thermal/sodimm<index>_temp_min_alarm`

Description	Read SODIMM temperature alarm status (critical, maximum, minimum). Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).
Values	index, Integer: 1-2 (N6100_LD, SN6600_LD); Alarm, Integer: 0 (clear), 1 (alarm)
Access	Read only
Release version	V.7.0060.1000
Example	Read SODIMM temperature alarms: `cat $bsp_path/thermal/sodimm1_temp_crit_alarm cat $bsp_path/thermal/sodimm1_temp_max_alarm cat $bsp_path/thermal/sodimm1_temp_min_alarm cat $bsp_path/thermal/sodimm2_temp_crit_alarm cat $bsp_path/thermal/sodimm2_temp_max_alarm cat $bsp_path/thermal/sodimm2_temp_min_alarm`

SODIMM Temperature Hysteresis

Node name	`$bsp_path/thermal/sodimm<index>_temp_crit_hyst` `$bsp_path/thermal/sodimm<index>_temp_max_hyst`
Description	Read SODIMM temperature hysteresis values for critical and max thresholds. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).
Values	index, Integer: 1-2 (N6100_LD, SN6600_LD); Hysteresis, Integer: Value in millidegrees Celsius (m°C)
Access	Read only
Release version	V.7.0060.1000
Example	Read SODIMM temperature hysteresis: `cat $bsp_path/thermal/sodimm1_temp_crit_hyst cat $bsp_path/thermal/sodimm1_temp_max_hyst cat $bsp_path/thermal/sodimm2_temp_crit_hyst cat $bsp_path/thermal/sodimm2_temp_max_hyst`

SWB ASIC Temperature

Node name	`$bsp_path/thermal/swb_asic<index>`
Description	Read switch-board (SWB) ASIC temperature when present on multi-board systems.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/swb_asic1`

Drive Temperature

Node name	`$bsp_path/thermal/drivetemp`
Description	Read NVMe/drive temperature from the `drivetemp` hwmon driver when linked by `hw-management-thermal-events.sh`.
Values	temperature, Integer: millidegrees Celsius
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/drivetemp`

Drive Critical Temperature

Node name	`$bsp_path/thermal/drivetemp_crit`
Description	Read drive critical temperature threshold.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/drivetemp_crit`

Drive Maximum Temperature Threshold

Node name	`$bsp_path/thermal/drivetemp_max`
Description	Read drive maximum temperature threshold.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/drivetemp_max`

Drive Minimum Temperature Threshold

Node name	`$bsp_path/thermal/drivetemp_min`
Description	Read drive minimum temperature threshold.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/thermal/drivetemp_min`

Watchdog

Note: On SN6600_LD the validated hierarchy places attributes under `$bsp_path/watchdog/main/` and `$bsp_path/watchdog/aux/` instead of directly under `$bsp_path/watchdog/`.

Read Boot Status

Node name	`$bsp_path/watchdog/bootstatus`
Description	Read watchdog boot status
Values	status, Integer
Access	Read only
Release version	1.0
Example	Read watchdog boot status: `cat $bsp_path/watchdog/bootstatus`

Read Identity

Node name	`$bsp_path/watchdog/identity`
Description	Read watchdog identity
Values	identity, String
Access	Read only
Release version	1.0
Example	Read watchdog identity: `cat $bsp_path/watchdog/identity`

Read No Way Out

Node name	`$bsp_path/watchdog/nowayout`
Description	Read watchdog no way out setting
Values	nowayout, Integer: 0/1
Access	Read only
Release version	1.0
Example	Read watchdog no way out: `cat $bsp_path/watchdog/nowayout`

Read State

Node name	`$bsp_path/watchdog/state`
Description	Read watchdog state
Values	state, String
Access	Read only
Release version	1.0
Example	Read watchdog state: `cat $bsp_path/watchdog/state`

Read Status

Node name	`$bsp_path/watchdog/status`
Description	Read watchdog status
Values	status, String
Access	Read only
Release version	1.0
Example	Read watchdog status: `cat $bsp_path/watchdog/status`

Read Timeout

Node name	`$bsp_path/watchdog/timeout`
Description	Read watchdog timeout value
Values	timeout, Integer: seconds
Access	Read only
Release version	1.0
Example	Read watchdog timeout: `cat $bsp_path/watchdog/timeout`

Read Timeleft

Node name	`$bsp_path/watchdog/timeleft`
Description	Read watchdog time left
Values	timeleft, Integer: seconds
Access	Read only
Release version	1.0
Example	Read watchdog time left: `cat $bsp_path/watchdog/timeleft`

BMC status

Sections BMC present and BMC to CPU control describe **host-CPU** views of BMC presence and I2C ownership (`hw-management.sh` / platform GPIO and mlxreg-io). MCTP address, MCTP bus and MCTP ready are **host** config/status for MCTP setup where the platform defines it (`hw-management.sh`).

On the **BMC** image, `bmc_to_cpu_ctrl` also appears under `$bsp_path/system/` via the **mlxreg-io** `regio` handler (see `bmc/examples/hw-management-bmc-system-sysfs.txt`).

BMC present

Stack	Host
Node name	`$bsp_path/system/bmc_present`
Description	GPIO indicating whether a BMC is present. Created by host `hw-management.sh` (platform-specific GPIO mapping).
Access	Read only
Release version	1.0
Example	`cat $bsp_path/system/bmc_present`

BMC to CPU control

Stack	Host (mlxreg-io on CPU); also **BMC** (`system/` symlink from `regio` on BMC image)
Node name	`$bsp_path/system/bmc_to_cpu_ctrl`
Description	I2C bus ownership between BMC and CPU (0 = CPU, 1 = BMC on supported platforms). On the host image the symlink targets **mlxplat** mlxreg-io; thermal control may read this node (`hw_management_thermal_control.py`: `system/bmc_to_cpu_ctrl`).
Access	Read/Write (platform-dependent)
Release version	1.0
Example	`cat $bsp_path/system/bmc_to_cpu_ctrl`

MCTP address

Stack	Host (`hw-management.sh` writes when platform defines MCTP, e.g. N5110)
Node name	`$bsp_path/config/mctp_addr`
Description	MCTP I2C client address used during host init to instantiate the MCTP device on `mctp_bus`.
Access	Read only (after init)
Release version	1.0
Example	`cat $bsp_path/config/mctp_addr`

MCTP bus

Stack	Host
Node name	`$bsp_path/config/mctp_bus`
Description	I2C bus number for MCTP client binding during host init.
Access	Read only (after init)
Release version	1.0
Example	`cat $bsp_path/config/mctp_bus`

MCTP ready

Stack	Host (mlxreg-io); **BMC** register `cpu_mctp_ready` under `$bsp_path/system/` on BMC image
Node name	`$bsp_path/system/cpu_mctp_ready`
Description	Indicates CPU MCTP readiness from platform mlxreg-io. Present in validated host trees (SN6600_LD, SN5810_LD, N6100_LD). On the BMC image the same logical register is listed in `nvsw_bmc_hid189_regio_data` / `hw-management-bmc-system-sysfs.txt`.
Access	Read only
Release version	1.0
Example	`cat $bsp_path/system/cpu_mctp_ready`

BMC reset cause

Stack: BMC (`hw-management-bmc` package on the BMC SoC)

The BMC reset-cause exporter (`hw-management-bmc-get-reset-cause.sh`) writes a separate subtree under **`$bsp_path/bmc/`**. This is distinct from host CPLD reset causes documented in Get Reset Cause (`$bsp_path/system/reset_*` on the switch CPU).

Source priority for SCU register words: U-Boot environment (`fw_printenv`), then `/proc/cmdline`, then `devmem`. Operator summary: `hw-management-bmc-show-reset-cause.sh` (see `bmc/README.md`).

Validation source: `bmc/usr/usr/bin/hw-management-bmc-get-reset-cause.sh`, `bmc/examples/hw-management-bmc-system-sysfs.txt` (host-side reset attrs only).

Primary BMC reset cause

Stack: BMC

Exactly **one** of the following files reads **1**; the others read **0** (v2 primary heuristic for HI189 / AST2700).

Node name	Description
`$bsp_path/bmc/reset_pwr_cycle`	Power-cycle-like BMC boot (no WDT log; SCU0 EXTRST# set)
`$bsp_path/bmc/reset_soft_reboot`	Warm reboot (WDT logged, or no WDT and EXTRST# clear)
`$bsp_path/bmc/reset_unknown`	WDT log and EXTRST# both set (conflicting / sticky path)

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/bmc/reset_pwr_cycle
cat $bsp_path/bmc/reset_soft_reboot
cat $bsp_path/bmc/reset_unknown
```

BMC reset domain detail

Stack: BMC

Hardware domain flags decoded from AST2700 SCU reset-event logs. Each file reads **0** or **1**.

Node name	Description
`$bsp_path/bmc/domains/reset_power_on`	PWRST indication (SCU0/SCU1)
`$bsp_path/bmc/domains/reset_external`	EXTRST# / SRST evidence
`$bsp_path/bmc/domains/reset_watchdog`	Non-software WDT bits or SCU0 watchdog evidence
`$bsp_path/bmc/domains/reset_software`	Software WDT bits in SCU1 0x080
`$bsp_path/bmc/domains/reset_cpu`	CPU domain reset
`$bsp_path/bmc/domains/reset_soc`	SoC domain reset
`$bsp_path/bmc/domains/reset_ahb`	AHB domain reset
`$bsp_path/bmc/domains/reset_caliptra`	Caliptra domain reset
`$bsp_path/bmc/domains/reset_usb`	USB subsystem reset
`$bsp_path/bmc/domains/reset_spi`	SPI domain reset
`$bsp_path/bmc/domains/reset_espi`	eSPI domain reset
`$bsp_path/bmc/domains/reset_emmc`	eMMC reset (SCU0)
`$bsp_path/bmc/domains/reset_msi`	MSI reset (SCU0)
`$bsp_path/bmc/domains/reset_security_watchdog2`	WDT2 nibble in SCU1 0x080

`$bsp_path/bmc/domains/reset_others`	No PWRST/WDT/CPU/WDT2 bits set

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/bmc/domains/reset_watchdog
```

Raw SCU reset event logs

Stack: BMC

Raw 32-bit SCU reset-event register values (hex text), written at export time.

Node name	SCU source (AST2700)
`$bsp_path/bmc/raw_scu0_reset_event_log0`	SCU0 0x050
`$bsp_path/bmc/raw_scu0_reset_event_log2`	SCU0 0x070
`$bsp_path/bmc/raw_scu1_reset_event_log0`	SCU1 0x050
`$bsp_path/bmc/raw_scu1_reset_event_log3`	SCU1 0x080

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/bmc/raw_scu0_reset_event_log0
```

BMC leakage A2D tree

Stack: BMC

Runtime leak-detector hierarchy under **$bsp_path/leakage/**, populated by `hw-management-bmc-a2d-leakage-config.sh` from `/etc/hw-management-bmc-a2d-leakage-config.json` (see `bmc/examples/hw-management-bmc-a2d-leakage-config-example.json`). Supported device types in the example config include **MAX1363**, **ADS1015**, and **ADS7924**. Udev events are handled by `hw-management-bmc-leakage-handler.sh`.

Layout reference: **bmc/examples/hw-management-bmc-leakage-sysfs.txt**.

N = leak-detector index (1-based order in the JSON array). **j** = channel index; channel directory names follow hardware `Channels[].Id` and may be non-contiguous (for example …/1/ and …/4/ only).

This tree is separate from host `$bsp_path/system/leakage<N>` mlxreg-io status (Leakage Sensor) and from `$bsp_path/events/leakage<N>` on the CPU image.

Leak detector attributes

Stack: BMC

Node name	Description
`$bsp_path/leakage/<N>/device_type`	Device type from JSON (for example ADS7924)
`$bsp_path/leakage/<N>/device_name`	Detector name from JSON `Name` field

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/leakage/1/device_type
cat $bsp_path/leakage/1/device_name
```

Leak channel attributes

Stack: BMC

Per-channel directory: `$bsp_path/leakage/<N>/<j>/`

Node name	Description
`input`	Raw IIO sample (symlink or file when `Probe` is true in JSON)
`min`	Low threshold (register-derived × scale, config units)
`max`	High threshold (register-derived × scale)
`warn`	WarningMax for this channel's `Type`
`crit`	CriticalMax for this channel's `Type`
`lwarn`	Optional WarningMin for this channel's `Type`
`lcrit`	Optional CriticalMin for this channel's `Type`
`type`	Channel type from JSON: `rop`, `flex`, or `embedded`
`scale`	Scale factor from JSON
`channel_name`	Optional human-readable name from JSON `ChnlNames`
`last_sample`	Optional 12-bit-aligned code (written on OOB event by leakage handler)
`last_event`	Optional event timestamp in milliseconds (written by leakage handler)

Access: Read only (handler may write `last_sample` / `last_event` on out-of-band events)

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/leakage/1/1/input
cat $bsp_path/leakage/1/1/warn
cat $bsp_path/leakage/1/1/type
```

Note: On BMC images for HI189, mlxreg-hotplug also exposes aggregate status registers such as `$bsp_path/system/leakage1`, `$bsp_path/system/leakage2`, and `$bsp_path/system/leakage_aggr` (see `bmc/examples/hw-management-bmc-system-sysfs.txt`).

JTAG interface

Enable / Disable JTAG mechanism

Node name	`$bsp_path/jtag/enable`		
Description	Enable or disable JTAG mechanism		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	enable	Integer	0/1
Example	Enable JTAG mechanism: `echo 1 > $bsp_path/jtag/enable`		

Set JTAG TCK pin

Node name	`$bsp_path/jtag/tck`		
Description	Set JTAG TCK pin		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	tck	Integer	
Example	Set JTAG TCK pin: `echo 1 > $bsp_path/jtag/tck`		

Set JTAG TDI pin

Node name	`$bsp_path/jtag/tdi`		
Description	Set JTAG TDI pin		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	tdi	Integer	
Example	Set JTAG TDI pin: `echo 1 > $bsp_path/jtag/tdi`		

Set JTAG TMS pin

Node name	`$bsp_path/jtag/tms`		
Description	Set JTAG TMS pin		
Access	Write		
Release version	1.0		
Arguments	**Name**	**Data type**	**Values**
	tms	Integer	
Example	Set JTAG TMS pin: `echo 1 > $bsp_path/jtag/tms`		

Get JTAG TDO pin

Node name	`$bsp_path/jtag/tdo`
Description	Get JTAG TDO pin
Values	tdo, Integer
Access	Read only
Release version	1.0
Example	Get JTAG TDO pin: `cat $bsp_path/jtag/tdo`

Thermal Control

The thermal algorithm is described in a separate document - Thermal Monitoring for NVIDIA Systems with Third Party OS

Thermal Control Versions

The hw-management package supports two versions of thermal control:

TC v2.0 (Default)

- **Script**: `/usr/bin/hw_management_thermal_control.py`
- **Service**: `hw-management-tc.service`
- **Description**: Used when `tc_version` in `tc_config.json` is missing or is anything other than `2.5` or `2.5.*`

TC v2.5

- **Script**: `/usr/bin/hw_management_thermal_control_2_5.py` (script `VERSION` is `2.5.1`)
- **Service**: `hw-management-tc.service`
- **Description**: `hw-management.sh` copies the platform thermal JSON to `$config_path/tc_config.json`. `hw-management-start-post.sh` reads `tc_version` from that file. `2.5` and `2.5.*` rewrite the service unit to this script. Any other value keeps the 2.0 script. If the platform sets no thermal config, the copy is `tc_config_not_supported.json`, whose `tc_version` is `2.0`.

Platforms on this branch whose copied config has `"tc_version": "2.5"`:

Platform	SKU	Config copied to `tc_config.json`	Config `name`
SN5600	HI144	`tc_config_msn5600.json`	`msn5600`
SN2201	HI168 only	`tc_config_msn2201_busbar.json`	`msn2201m`
MSN4700 DGX	HI184	`tc_config_msn4700d.json`	`msn4700d`
Q3401	HI179	`tc_config_q3401.json`	`q3401_rd`
SN5610	HI172	`tc_config_sn5610.json`	`sn5610`
SN5640	HI171	`tc_config_sn5640.json`	`sn5640`

`sn5640_specific()` (board `VMOD0022`) uses `tc_config_sn5640.json` for every SKU other than HI172. SN5400 (HI147) uses `tc_config_msn5400.json` (`2.0`). SN5600d (HI174) uses `tc_config_msn5600d.json` (`2.0`). SN5700 (HI148) sets no thermal config, so it gets `tc_config_not_supported.json` (`2.0`). SN2201 SKUs other than HI168 use `tc_config_msn2201.json` (`2.0`).

Drivers

`$bsp_path` is the NOS-facing tree. hw-management builds it with symlinks and small files on top of the kernel modules below. The attribute sections are the interface to use. Each

section below is the behavior of that module.

mp2855 is added by the hw-mgmt Linux 6.12 patch recipes-kernel/linux/linux-6.12/0043-hwmon-pmbus-Add-support-for-MPS-Multi-phase-mp2855-c.patch. The other modules are the drivers named in each section.

mlx-platform

Kernel module mlx_platform (drivers/platform/mellanox/mlx-platform.c). MODULE_DESCRIPTION is “Mellanox platform driver”. The platform device name is mlxplat.

The driver matches the board and registers the CPLD register map and the child devices the other modules in this chapter bind to: I2C controller, I2C mux, regmap I/O, hotplug, LEDs, fan, and watchdog. Watchdog platform data from this driver supplies the register set, the identity (mlx-wdt-main or mlx-wdt-aux), the initial timeout, the action register, and the nowayout and start_at_boot feature bits.

mlxreg-hotplug

Kernel module mlxreg-hotplug (drivers/platform/mellanox/mlxreg-hotplug.c). MODULE_DESCRIPTION is “Mellanox regmap hotplug platform driver”.

The driver takes the platform interrupt, caches the aggregation register, and walks the items in its platform data. The work handler acts on fan insertion and removal, PSU insertion and removal, power-cable attach and detach, and ASIC health. ASIC good health is MLXREG_HOTPLUG_GOOD_HEALTH_MASK (0x02): 0x02 is good health, 0x00 is bad or dormant, and 0x03 is booting. The driver allocates at most MLXREG_HOTPLUG_ATTRS_MAX (128) hwmon attributes. Which items exist is the platform data from mlx-platform. hw-management turns those events into files under $bsp_path/events/.

mlxreg-io

Kernel module mlxreg-io (drivers/platform/mellanox/mlxreg-io.c). MODULE_DESCRIPTION is “Mellanox regmap I/O access driver”.

Probe requires platform data. `mlxreg_io_attr_init()` creates one sysfs attribute per entry: the name is `data[i].label` and the mode is `data[i].mode`. Those attributes are registered on an hwmon device named `mlxreg_io`. On these systems the labels are the reset, reset-cause, mux, and other registers that `mlx-platform` passes in. hw-management symlinks the ones this manual documents into `$bsp_path/system/`, `$bsp_path/config/`, and the line-card and DPU trees.

mlxreg-fan

Kernel module `mlxreg-fan` (`drivers/hwmon/mlxreg-fan.c`). `MODULE_DESCRIPTION` is "Mellanox FAN driver".

The driver reads PWM and tachometer registers from platform data and registers one hwmon device. The hwmon channel table has 24 tachometers (`MLXREG_FAN_MAX_TACHO`) and 4 PWM outputs (`MLXREG_FAN_MAX_PWM`). A channel is visible only when platform data marks it connected. Tachometer entries are the platform-data labels that contain `tacho`.

hwmon files for each connected channel:

- `fan[1-24]_input` — tachometer, read only
- `fan[1-24]_fault` — tachometer fault, read only
- `pwm[1-4]` — duty cycle, read/write

When `CONFIG_THERMAL` is built in, each connected PWM gets a cooling device named `mlxreg_fan`, `mlxreg_fan1`, `mlxreg_fan2`, or `mlxreg_fan3`. `cur_state` is read/write. `max_state` is read only. RPM uses the driver formula `15000000 * 100 / ((regval + samples) * divider)`, with default samples per pulse `44`. hw-management publishes the linked nodes under `$bsp_path/thermal/`.

mlxreg-lc

Kernel module `mlxreg-lc` (`drivers/platform/mellanox/mlxreg-lc.c`). `MODULE_DESCRIPTION` is "Nvidia line card platform driver".

The driver defines one line-card type, `MLXREG_LC_SN4800_C16` (`0x0000`): a 100GbE line card with 16 QSFP28 ports. Probe reads the type from `MLXREG_LC_REG_CONFIG_OFFSET` and accepts only that value. It maps the CPLD and FPGA version, part number, and minor

version, plus reset cause, LED, maximum power, and configuration registers, and registers an `i2c-mux-mlxcpld` device, an `mlxreg-io` device, and a `leds-mlxreg` device. `mlxreg-hotplug` calls back on line-card events. On a sync or power event the driver enables the card when the state already includes `MLXREG_LC_CONFIGURED` (`MLXREG_LC_INITIALIZED | MLXREG_LC_POWERED | MLXREG_LC_SYNCED`). The NOS-facing files are the line-card sections of this manual.

mlxreg-dpu

Kernel module `mlxreg-dpu` (`drivers/platform/mellanox/mlxreg-dpu.c`). `MODULE_DESCRIPTION` is "Nvidia Data Processor Unit platform driver".

The driver defines one type, `MLXREG_DPU_BF3` (`0x0050`): a DPU with a BF3 SoC. Probe builds an I2C regmap and registers an `mlxreg-io` device and an `mlxreg-hotplug` device. The I/O labels in the driver are `fpga1_version`, `fpga1_pn`, `fpga1_version_min`, `perst_rst`, `usbphy_rst`, `phy_rst`, `tpm_rst`, `reset_from_main_board`, `reset_aux_pwr_or_reload`, `reset_comex_pwr_fail`, `reset_dpu_thermal`, `reset_pwr_off`, `dpu_id`, `voltreg_update_status`, `boot_progress`, and `ufm_upgrade`. Hotplug labels are the power-good inputs (`pg_vddio`, `pg_vdd_cpu`, `pg_vdd`, `pg_1v8`, `pg_comparator`, `pg_vddq`, `pg_hvdd`, `pg_dvdd`) and the health events (`thermal_trip`, `ufm_upgrade_done`, `vddq_hot_alert`, `vdd_cpu_hot_alert`, `vddq_alert`, `vdd_cpu_alert`). The NOS-facing files are the DPU sections of this manual.

mlxsw

Kernel module `mlxsw_core` (`drivers/net/ethernet/mellanox/mlxsw/core.c`). `MODULE_DESCRIPTION` is "Mellanox switch device core driver".

`drivers/net/ethernet/mellanox/mlxsw/core_hwmon.c` registers the hwmon device with the name `mlxsw`. That name is what hw-management tests before it writes `$bsp_path/sysfs_mlxsw_rdy`. See mlxsw sysfs ready.

`mlxsw_spectrum` (`MODULE_DESCRIPTION` "Mellanox Spectrum driver") binds four PCI IDs: `PCI_DEVICE_ID_MELLANOX_SPECTRUM`, `SPECTRUM2`, `SPECTRUM3`, and `SPECTRUM4`, on the drivers `mlxsw_sp1` through `mlxsw_sp4`.

mlxsw-i2c

Kernel module `mlxsw_i2c` (`drivers/net/ethernet/mellanox/mlxsw/i2c.c`). `MODULE_DESCRIPTION` is “Mellanox switch I2C interface driver”.

The driver registers an `mlxsw_bus` with kind `"i2c"`. That bus implements init, fini, skb transmit, and command execution (`mlxsw_i2c_cmd_exec`). `mlxsw_minimal` registers its I2C client driver through `mlxsw_i2c_driver_register()`.

mlxsw-minimal

Kernel module `mlxsw_minimal` (`drivers/net/ethernet/mellanox/mlxsw/minimal.c`). `MODULE_DESCRIPTION` is “Mellanox minimal driver”.

The I2C id and driver name are `mlxsw_minimal`. The module registers that I2C driver on the `mlxsw_i2c` bus. The hwmon device it uses is still registered as `mlxsw` by `core_hwmon.c`. On SPC1, hw-management creates the module, ASIC, and fan links from that hwmon and then writes `$bsp_path/sysfs_mlxsw_rdy`.

i2c-mlxcpld

Kernel module `i2c-mlxcpld` (`drivers/i2c/busses/i2c-mlxcpld.c`). `MODULE_DESCRIPTION` is “Mellanox I2C-CPLD controller driver”.

The driver registers one numbered I2C adapter (`i2c_mlxcpld`) and runs each transfer by polling the CPLD status register. The poll interval is `MLXCPLD_I2C_POLL_TIME` (200 microseconds). The transfer timeout is `MLXCPLD_I2C_XFER_TO` (500 ms). The data-register block is 36 bytes (`MLXCPLD_I2C_DATA_REG_SZ`). The capability register selects that size, an extended size of twice the block, or a second extended size of `(36 - 4) * 4`, and bit 7 of that register enables SMBus block read.

i2c-mux-mlxcpld

Kernel module `i2c-mux-mlxcpld` (`drivers/i2c/muxes/i2c-mux-mlxcpld.c`). `MODULE_DESCRIPTION` is “Mellanox I2C-CPLD-MUX driver”.

The CPLD I2C controller has one physical bus. This driver writes a CPLD mux-select register to connect that bus to one downstream channel and creates a virtual I2C adapter per channel. The select register is 1 or 2 bytes, from platform data. The mux write does not call `i2c_transfer()` or `i2c_smbus_xfer()`, because those would lock the adapter a second time. It uses `__i2c_smbus_xfer()` for a 1-byte register and a raw `i2c_msg` for a 2-byte register.

leds-mlxreg

Kernel module `leds-mlxreg` (`drivers/leds/leds-mlxreg.c`). `MODULE_DESCRIPTION` is “Mellanox LED regmap driver”.

Each LED is one nibble of a CPLD register, accessed through regmap, and is registered as an LED class device. Codes in the driver: off `0x00`, solid red `0x05`, solid amber `0x09`, solid green `0x0d`. Blink is an offset from the solid code: `0x01` for 3 Hz (about 167 ms) and `0x02` for 6 Hz (about 83 ms). The NOS-facing color and control files are the LED section of this manual and `LED_Control_API.md`.

PMBus and hwmon

These drivers publish the standard hwmon attributes (voltage, current, power, temperature, fan). hw-management links the attributes named in the platform sensor configuration into `$bsp_path/environment/` and `$bsp_path/thermal/`.

mp2975

Kernel module `mp2975` (`drivers/hwmon/pmbus/mp2975.c`). `MODULE_DESCRIPTION` is “PMBus driver for MPS MP2975 device”.

The same module probes mp2975, mp2973, and mp2971. Maximum phases are 8 and 4 for MP2975, 14 and 6 for MP2973, and 8 and 3 for MP2971 (mp2975_max_phases). Probe starts at one page and sets pages to 2 (MP2975_PAGE_NUM) when rail 2 is present. Phase counts and the VID mode are read from the device. mp2975_identify_vid() sets the rail to IMVP9, VR12, or VR13. For MP2975 the driver info formats are linear for input voltage and input current, and direct for output voltage, output current, temperature, and power. MP2973 and MP2971 use direct format for output voltage and linear format for input voltage, both currents, temperature, and power. Page 0 reports input voltage, output voltage, input current, output current, input power, output power, temperature, and the matching status bits. Rail 2 receives MP2975_RAIL2_FUNC when it is connected.

mp2888

Kernel module mp2888 (drivers/hwmon/pmbus/mp2888.c). MODULE_DESCRIPTION is “PMBus driver for MPS MP2888 device”.

The driver info has one page. MP2888_MAX_PHASE is 10. Formats are linear for input voltage and input current, and direct for output voltage, output current, temperature, and power. Output voltage mode is direct (PB_VOUT_MODE_DIRECT).

mp2891

Kernel module mp2891 (drivers/hwmon/pmbus/mp2891.c). MODULE_DESCRIPTION is “PMBus driver for MPS MP2891”.

The driver uses two pages (MP2891_PAGE_NUM). The driver info sets direct format for input voltage, input current, output current, temperature, and power, and forces VOUT to direct format. Registers that the device returns as linear11 (input power, output power, and the related estimates) are converted with mp2891_reg2data_linear11() before they are reported. Probe stores a VOUT scale and an IOUT scale per rail.

mp2855

hw-mgmt adds drivers/hwmon/pmbus/mp2855.c in the Linux 6.12 kernel patch series. MODULE_DESCRIPTION is “PMBus driver for MPS MP2855 device”. It is the MPS MP2855

dual-loop digital multi-phase controller used on an AMD SVI2 2.0 platform, with up to 9 phases on rail 1 and up to 4 phases on rail 2. The module name is `mp2855`.

The driver uses two pages (`MP2855_PAGE_NUM` is 2). `MP2855_RAIL1_FUNC` includes input voltage, output voltage, temperature, output current, and the matching status bits. `MP2855_RAIL2_FUNC` includes output voltage, temperature, output current, and status, and does not include input voltage. Probe reads a VOUT scale and an IOUT scale for each rail.

isl68137

Kernel module `isl68137` (`drivers/hwmon/pmbus/isl68137.c`). `MODULE_DESCRIPTION` is "PMBus driver for Renesas digital multiphase voltage regulators".

The I2C id table `raa_dmpvr_id` binds these parts: `isl68137`; `isl68220` through `isl68227`, `isl68229`, `isl68233`, `isl68239`; `isl69222` through `isl69225`, `isl69227`, `isl69228`, `isl69234`, `isl69236`, `isl69239`, `isl69242`, `isl69243`, `isl69247`, `isl69248`, `isl69254` through `isl69256`, `isl69259`, `isl69260`, `isl69268`, `isl69269`, `isl69298`; `raa228000`, `raa228004`, `raa228006`, `raa228228`, `raa228244`, `raa228246`, `raa228942`, `raa228943`, `raa229001`, `raa229004`, `raa229141`, and `raa229621`.

Page 0 of the driver info exposes input voltage, input current, input power, status input, temperature (including temp2 and temp3), status temperature, output voltage, status output voltage, output current, status output current, output power, and VMON. Some variants clear the VMON bit or the temp3 bit.

powr1220

Kernel module `powr1220` (`drivers/hwmon/powr1220.c`). `MODULE_DESCRIPTION` is "POWR1220 driver".

The driver reads the ADC voltage inputs of the Lattice POWR1220 and POWR1014 (`enum powr1xxx_chips`). Probe sets `max_channels` to 12 for POWR1220 and to 10 for POWR1014. The conversion is a 12-bit value times `ADC_STEP_MV` (2 mV). When the stored maximum for a channel is above `ADC_MAX_LOW_MEASUREMENT_MV` (2000 mV), or there is no maximum yet, `powr1220_read_adc()` sets the attenuator bit in `ADC_MUX`. hwmon names are `in0` upward, mapped to the VMON pins.

emc2305

Kernel module `emc2305` (`drivers/hwmon/emc2305.c`). `MODULE_DESCRIPTION` is "Microchip EMC2305 fan controller driver".

The I2C ids are `emc2301`, `emc2302`, `emc2303`, and `emc2305`. Probe reads the product id and sets the PWM count: EMC2301 has 1, EMC2302 has 2, EMC2303 has 3, and EMC2305 has 5 (`EMC2305_PWM_MAX` is 5). PWM duty is 0 through `EMC2305_FAN_MAX` (`0xff`). The default cooling `max_state` is `EMC2305_FAN_MAX_STATE` (10). Tachometer counts are read from the per-fan tach registers and converted with `EMC2305_RPM_FACTOR` (3932160). hwmon files follow the PWM count: `fan[1-5]_input` and `fan[1-5]_fault` are read only, and `pwm[1-5]` is read/write. Platform data `pwm_separate` gives each channel its own PWM and cooling device. When `pwm_separate` is clear, every PWM is bound to one thermal zone and runs at the same speed. Cooling files are `cur_state` and `max_state`.

lm25066

Kernel module `lm25066` (`drivers/hwmon/pmbus/lm25066.c`). `MODULE_DESCRIPTION` is "PMBus driver for LM25066 and compatible chips".

The I2C ids are `lm25056`, `lm25066`, `lm5064`, `lm5066`, and `lm5066i`. hw-management uses the `lm5066i` id for the PDB hot-swap controller; its power and current scale factors are the `pdb_hotswap<N>_power1_scale` and `_curr1_scale` nodes in Environment Control. The module is listed in `modules-load.d`.

tps53679

Kernel module `tps53679` (`drivers/hwmon/pmbus/tps53679.c`). `MODULE_DESCRIPTION` is "PMBus driver for Texas Instruments TPS53679".

The I2C ids are `bmr474`, `tps53647`, `tps53667`, `tps53676`, `tps53679`, `tps53681`, and `tps53688`. The module is listed in `modules-load.d`, and hw-management instantiates it from the platform connect tables.

xdpe12284

Kernel module `xdpe12284` (`drivers/hwmon/pmbus/xdpe12284.c`). `MODULE_DESCRIPTION` is "PMBus driver for Infineon XDPE122 family".

Upstream ids are `xdpe11280`, `xdpe12254`, and `xdpe12284`. The hw-mgmt Linux 6.12 patch `0019-Extend-driver-to-support-Infineon-Digital-Multi-phas.patch` adds `xdpe15284` and `xdpe1a2g7` (and the matching `infineon,` device-tree compatibles). The platform tables in `hw-management-devtree.sh` use `xdpe12284` and `xdpe15284` for the COMEX voltage monitors and `xdpe1a2g7` for the board voltage monitors that name it.

xdpe1a2g7b

hw-mgmt adds `drivers/hwmon/pmbus/xdpe1a2g7b.c` in the Linux 6.12 kernel patch `0059-hwmon-pmbus-xdpe1a2g7b-Add-support-for-XDPE1A2G7B-5B.patch`. `MODULE_DESCRIPTION` is "PMBus driver for Infineon XDPE1A2G7B/5B". It is enabled by `CONFIG_SENSORS_XDPE1A2G7B`.

mp2845

hw-mgmt adds `drivers/hwmon/pmbus/mp2845.c` in the Linux 6.12 kernel patch `0056-hwmon-pmbus-mp2845-Add-support-for-MP2845-device.patch`. `MODULE_DESCRIPTION` is "MP2845 driver". The I2C id is `mp2845`. The module is listed in `modules-load.d`.

mp2869

hw-mgmt adds `drivers/hwmon/pmbus/mp2869.c` in the Linux 6.12 kernel patch `0041-hwmon-add-MP2869-MP29608-MP29612-and-MP29816-series-.patch`. `MODULE_DESCRIPTION` is "PMBus driver for MPS MP2869". The I2C ids are `mp2869`, `mp29608`, `mp29612`, and `mp29816`. The `mp29816` id is the one `hw-management-devtree.sh` uses for the MPS voltage monitors (`mp29816`, `mp29816-6x`, `mp29816-7x` board variants).

mp29502

hw-mgmt adds `drivers/hwmon/pmbus/mp29502.c` in the Linux 6.12 kernel patch `0042-hwmon-add-MP29502-driver.patch`. `MODULE_DESCRIPTION` is "PMBus driver for MPS MP29502". The I2C id is `mp29502`.

mp5926

Kernel module `mp5926` (`drivers/hwmon/pmbus/mp5926.c`), added by the hw-mgmt patches `0108-hwmon-pmbus-mp5926-Add-support-for-MP5926-device.patch` (Linux 6.1) and `0057-hwmon-pmbus-mp5926-Add-support-for-MP5926-device.patch` (Linux 6.12). The I2C id is `mp5926`. The BMC BOM files instantiate it as the PDB hot-swap controller (spec `mp5926 0x12 <bus> pdb_hotswap<N>`). It does not create `config/pdb_hotswap_scale`; that node exists only for `lm5066i`.

Temperature sensors

hw-management links the attributes of these drivers into `$bsp_path/thermal/` and `$bsp_path/environment/` as the platform sensor configuration names them. Kernel version `6.12` is the reference.

lm75

Kernel module `lm75` (`drivers/hwmon/lm75.c`). `MODULE_DESCRIPTION` is "LM75 driver". One module binds 29 I2C ids. The ids the platform tables use are `adt75`, `tmp75`, `tmp1075`, and `lm75`; the full list is in the driver. The BMC stack instantiates `lm75` for the BMC board temperature (see Host and BMC software stacks).

tmp102

Kernel module `tmp102` (`drivers/hwmon/tmp102.c`). `MODULE_DESCRIPTION` is "Texas Instruments TMP102 temperature sensor driver". The I2C id is `tmp102`. The module is listed in `modules-load.d`.

tmp421

Kernel module `tmp421` (`drivers/hwmon/tmp421.c`). `MODULE_DESCRIPTION` is "Texas Instruments TMP421/422/423/441/442 temperature sensor driver". The I2C ids are `tmp421`, `tmp422`, `tmp423`, `tmp441`, and `tmp442`.

tmp401

Kernel module `tmp401` (`drivers/hwmon/tmp401.c`). `MODULE_DESCRIPTION` is "Texas Instruments TMP401 temperature sensor driver". The I2C ids are `tmp401`, `tmp411`, `tmp431`, `tmp432`, and `tmp435`. hw-management uses `tmp411` from the device tree table.

lm90

Kernel module `lm90` (`drivers/hwmon/lm90.c`). `MODULE_DESCRIPTION` is "LM90/ADM1032 driver". It binds 45 I2C ids including `lm90`, `adm1032`, `max6659`, and `tmp451`. hw-management uses `tmp451` from the platform tables.

emc1403

Kernel module `emc1403` (`drivers/hwmon/emc1403.c`). The I2C ids are `emc1402`, `emc1403`, `emc1404`, `emc1412`, `emc1413`, `emc1414`, `emc1422`, and `emc1423`. hw-management uses `emc1403` from the device tree table. The module is enabled by `CONFIG_SENSORS_EMC1403`.

stts751

Kernel module `stts751` (`drivers/hwmon/stts751.c`). `MODULE_DESCRIPTION` is "STTS751 sensor driver". The I2C id is `stts751`.

spd5118

Kernel module `spd5118` (`drivers/hwmon/spd5118.c`). `MODULE_DESCRIPTION` is "SPD 5118 driver". It reads the temperature sensor of DDR5 SPD hubs. The hw-mgmt Linux 6.1 patch `0120-hwmon-Add-support-for-SPD5118-compliant-temperature-.patch` backports it; 6.12 has it upstream. It is enabled by `CONFIG_SENSORS_SPD5118`.

coretemp

Kernel module `coretemp` (`drivers/hwmon/coretemp.c`). `MODULE_DESCRIPTION` is "Intel Core temperature monitor". `hw-management.sh` runs `modprobe coretemp` on every CPU type except AMD SNW, AMD V3000, AMD FRNG, and BF3, because the driver supports only Intel CPUs. It is also skipped on SimX.

jc42

Kernel module `jc42` (`drivers/hwmon/jc42.c`). The I2C id is `jc42`. It reads the temperature sensor of DDR4 SODIMMs. `hw-management-helpers.sh` runs `modprobe jc42` when `/sys/bus/i2c/drivers/jc42` is missing, writes the SODIMM critical limit to each bound sensor, and `hw-management.sh` instantiates `jc42` by hand on AMD platforms where the driver does not probe the sensor itself. DDR5 SPD hubs use `spd5118` instead.

sbtsi

I2C id `sbtsi` (the AMD SB-TSI CPU temperature interface). It exposes `temp1_input`, `temp1_max`, and `temp1_min`. Only the BMC stack uses it: the HI189 early-I2C map binds bus 15 / `0x4c` to it, and it is the source of `thermal/cpu_temp_input`, `cpu_temp`, and `cpu_min` (see Host and BMC software stacks).

drivetemp

Kernel module `drivetemp` (`drivers/hwmon/drivetemp.c`). It reports the temperature of SATA drives (it does not handle NVMe). `hw-management.sh` runs `modprobe drivetemp` only when the module file exists for the running kernel. `hw-management-thermal-events.sh`

links its temp1_* attributes to $bsp_path/thermal/drivetemp and the _crit, _max, and _min nodes. NVMe drives reach the same virtual nodes through a separate branch of that script: for hwmon devices named nvme it links the composite temperature (temp1_*) to the same drivetemp* nodes and each additional labelled sensor to drivetemp_<label>, where the script first removes the spaces from the label and lowercases it (for example, the label Sensor 1 becomes drivetemp_sensor1).

ADC, GPIO, EEPROM and I2C mux

max1363

Kernel module max1363 (drivers/iio/adc/max1363.c). MODULE_DESCRIPTION is "Maxim 1363 ADC". The id max11603 is the A2D device the platform tables place at address 0x6d (or 0x64) for the voltage and current inputs of older systems. The module is listed in modules-load.d.

ti-ads1015

Kernel module ti-ads1015 (drivers/iio/adc/ti-ads1015.c). MODULE_DESCRIPTION is "Texas Instruments ADS1015 ADC driver". The I2C ids are ads1015, ads1115, and tla2024. It reads the leak detector inputs in the BMC leakage A2D tree. The module is listed in modules-load.d.

ads7924

I2C id ads7924 (Texas Instruments ADS7924, 4-channel 12-bit ADC). The BMC leakage configuration script maps DeviceType ADS7924 to this driver name and binds it through new_device when Probe is true. Its alarm thresholds are programmed with raw I2C transfers (hw-management-bmc-ads7924-force-alarm.sh and hw-management-bmc-a2d-leakage-config.sh). It is one of the devices behind the BMC leakage A2D tree.

i2c-dev

Kernel module `i2c-dev`. It creates the `/dev/i2c-<N>` character devices and the `i2c-dev` class entries. It is listed in `modules-load.d`. The udev rules handle the `i2c-dev` subsystem events for the CPLD I2C adapters and call `hw-management-chassis-events.sh` with `i2c_link`, which creates the per-bus links.

ledtrig-timer

Kernel module `ledtrig-timer`. It provides the `timer` LED trigger, which blinks an LED at a configured rate, and it is listed in `modules-load.d`.

at24

Kernel module `at24` (`drivers/misc/eeprom/at24.c`). `MODULE_DESCRIPTION` is "Driver for most I2C EEPROMs". hw-management creates the nodes under `$bsp_path/eeprom/` on top of the `24c02`, `24c512`, and related ids. See EEPROM Control.

gpio-pca953x

Kernel module `gpio-pca953x` (`drivers/gpio/gpio-pca953x.c`). `MODULE_DESCRIPTION` is "GPIO expander driver for PCA953x". It binds 37 I2C ids, for example `pca9535`, `pca9555`, `pca9557`, and `tca9539`. It is listed in `modules-load.d` (arm64 list).

i2c-mux-pca954x

Kernel module `i2c-mux-pca954x` (`drivers/i2c/muxes/i2c-mux-pca954x.c`). `MODULE_DESCRIPTION` is "PCA954x I2C mux/switch driver". It binds 18 I2C ids, from `pca9540` to `pca9849` and the `max7356` to `max7369` family. Each downstream channel is a virtual I2C adapter. It is listed in `modules-load.d` (arm64 list).

i2c-mux-reg

Kernel module `i2c-mux-reg` (`drivers/i2c/muxes/i2c-mux-reg.c`). `MODULE_DESCRIPTION` is "Register-based I2C multiplexer driver". It selects the downstream channel by writing a platform-data register. It is listed in `modules-load.d` and enabled by `CONFIG_I2C_MUX_REG`.

Host I2C controllers

i2c-i801

Kernel module `i2c-i801` (`drivers/i2c/busses/i2c-i801.c`). `MODULE_DESCRIPTION` is "I801 SMBus driver". It is the SMBus controller of the Intel switch CPU PCH. It is listed in `modules-load.d`.

i2c-designware

Kernel module `i2c_designware_platform` (`drivers/i2c/busses/i2c-designware-platdrv.c`). `MODULE_DESCRIPTION` is "Synopsys DesignWare I2C bus adapter". `hw-management.sh` runs `modprobe i2c_designware_platform` for SKUs HI162, HI166, HI167, HI169, HI170, HI176, and HI177 (N51XX_LD), and on AMD SNW CPUs.

i2c-asf

hw-mgmt adds `drivers/i2c/busses/i2c-asf.c` in the Linux 6.12 kernel patch `0054-i2c-asf-Introduce-MCTP-support-over-ASF-controller.patch`. `MODULE_DESCRIPTION` is "ASF SMBus driver". `hw-management.sh` runs `modprobe i2c_asf` for SKUs HI162, HI166, HI167, HI169, HI170, HI176, and HI177 (N51XX_LD); see `config/mctp_bus` and `config/mctp_addr` in BMC status. It is enabled by `CONFIG_I2C_ASF`.

Other modules

mlxbf-ptm

hw-mgmt adds `drivers/platform/mellanox/mlxbf-ptm.c` in the Linux 6.12 patches `0049` to `0051` (and the 5.10 and 6.1 equivalents). `MODULE_DESCRIPTION` is "Nvidia Bluefield power and thermal debugfs driver". It is listed in the arm64 `modules-load.d` list, and enabled by `CONFIG_MLXBF_PTM`. It exposes debugfs only; it does not create `$bsp_path` nodes.

icp201xx

hw-mgmt adds the InvenSense ICP-201xx barometric pressure sensor driver (`drivers/iio/pressure/icp201xx*.c`) in `9003-iio-pressure-icp20100-OPT-add-driver-for-InvenSense-.patch`. `MODULE_DESCRIPTION` is "Invensense ICP201XX driver". The I2C id is `icp201xx`. The OPT-specific patch is built only for the images that carry it.

BMC and host helper modules

These are not hw-management drivers. Scripts load them by name: `ftgmac100` (`drivers/net/ethernet/faraday/ftgmac100.c`, "FTGMAC100 driver", the BMC Ethernet MAC, loaded by `hw-management-bmc-helpers-common.sh`), `g_ether` (`drivers/usb/gadget/legacy/ether.c`, USB Ethernet gadget, loaded by `hw-management-bmc-ready-common.sh`), and `nvidia_drm` (the NVIDIA GPU driver, loaded by `hw-management-start-post.sh`).

Watchdog (mlx-wdt)

Kernel module `mlx_wdt` (`drivers/watchdog/mlx_wdt.c`). `MODULE_DESCRIPTION` is "Mellanox watchdog driver". The NOS-facing files are `$bsp_path/watchdog/`. The driver is probed from `mlx_platform`. Platform data supplies the regmap, the identity, the initial timeout, and the feature bits.

The identity `mlx-wdt-main` selects `mlxreg_wdt_main_info`, whose options add `WDIOF_CARDRESET`. Any other identity selects `mlxreg_wdt_aux_info` (`mlx-wdt-aux`), whose options

add WDIOF_ALARMONLY. Access is regmap. Probe records regmap_get_val_bytes() as the register width.

enum mlxreg_wdt_type in include/linux/platform_data/mlxreg.h defines three types. The platform-data version selects the type:

- Type 1. Old systems. set_timeout stores order_base_2(timeout * 1000), so the hardware timeout is a power of two milliseconds. A 20 second request becomes 32768 ms, and the reported timeout is 32 seconds. Maximum is MLXREG_WDT_MAX_TIMEOUT_TYPE1 (32 seconds). The type 1 ops do not implement get_timeleft.
- Type 2. New systems. The timeout register is the requested value in seconds. Maximum is MLXREG_WDT_MAX_TIMEOUT_TYPE2 (255 seconds). get_timeleft reads one register.
- Type 3. Present on systems with the new CPLD, selected by the WD capability bit. The set-time and time-left registers are two bytes. Maximum is MLXREG_WDT_MAX_TIMEOUT_TYPE3 (65535 seconds). Type 3 uses the type 2 ops.

The action, timeout, time-left, ping, and reset registers are the platform-data entries whose labels contain action, timeout, timeleft, ping, and reset. MLXREG_CORE_WD_FEATURE_NOWAYOUT calls watchdog_set_nowayout(), so the watchdog cannot be stopped once started. MLXREG_CORE_WD_FEATURE_START_AT_BOOT starts the watchdog in probe and sets WDOG_HW_RUNNING. Probe then calls mlxreg_wdt_check_card_reset(), which reads the reset register and logs when the watchdog previously reset the CPU.